



深度学习平台与应用

第十讲：目标检测与图像分割

大纲

- 语义分割
- 目标检测
- 实例分割

■ 图像分类



假设有一个标签集合
{人, 狗, 猫, 汽车, ...}

模型的分类预测

猫

■ 计算机视觉任务

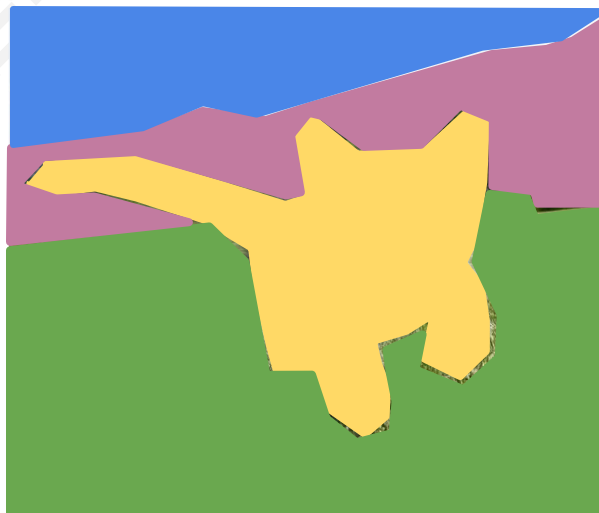
Classification



CAT

No spatial extent

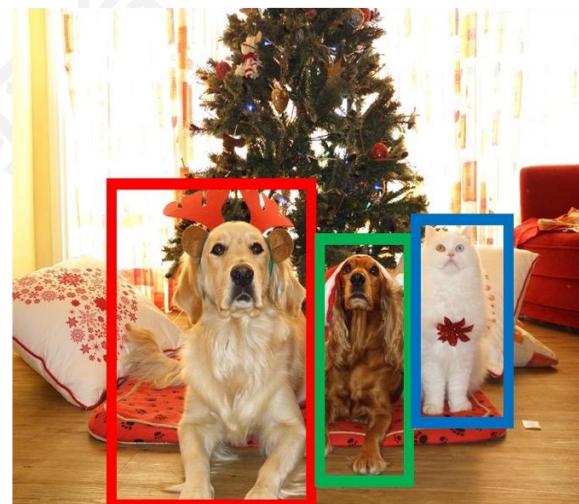
Semantic
Segmentation



GRASS, CAT, TREE,
SKY

No objects, just pixels

Object
Detection



DOG, DOG, CAT

Multiple Object

Instance
Segmentation



DOG, DOG, CAT

■ 语义分割-Semantic Segmentation

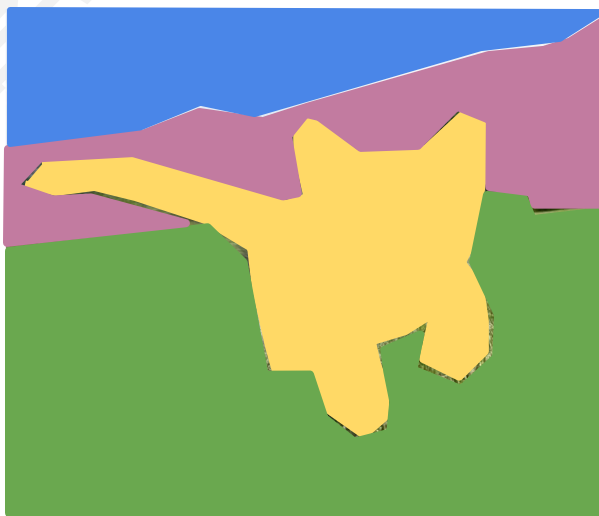
Classification



CAT

No spatial extent

Semantic
Segmentation



GRASS, CAT, TREE,
SKY

No objects, just pixels

Object
Detection



DOG, DOG, CAT

Multiple Object

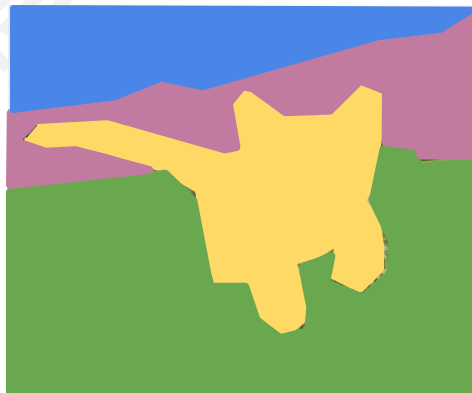
Instance
Segmentation



DOG, DOG, CAT

■ 语义分割

对图像的每个像素进行分类



GRASS, CAT, TREE, SKY, ...

训练



?

测试

■ 语义分割-滑窗计算



?

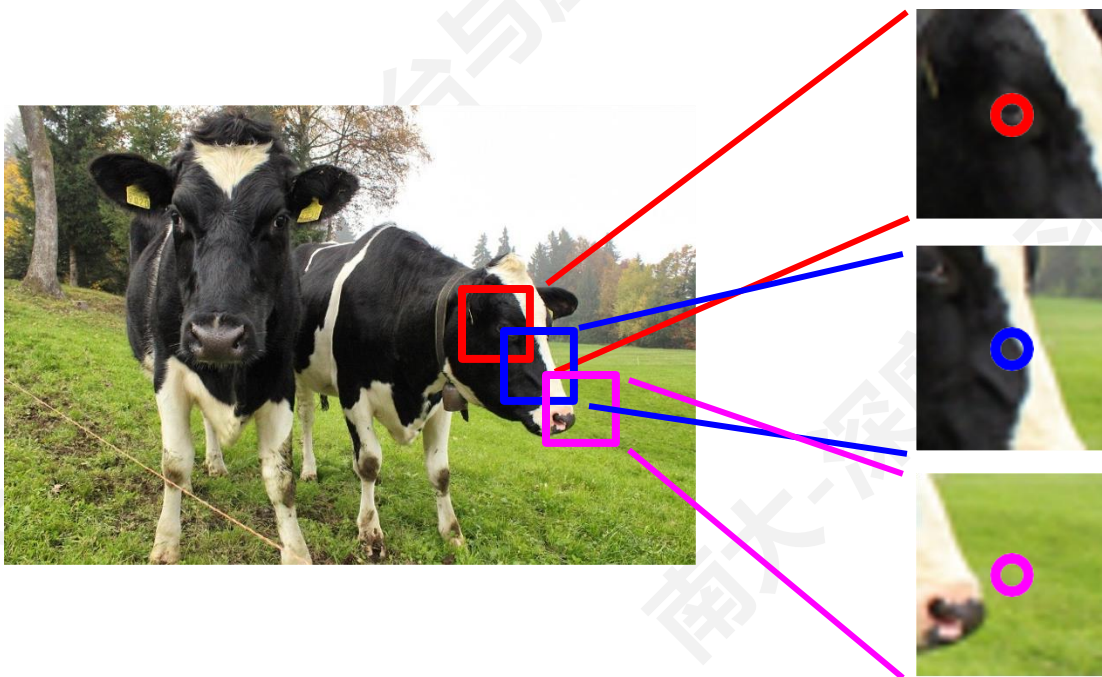
■ 语义分割-滑窗计算



直接对每个像素进行分类

**问题：如果没有语义信息，
很难正确分类每个像素**

■ 语义分割-滑窗计算

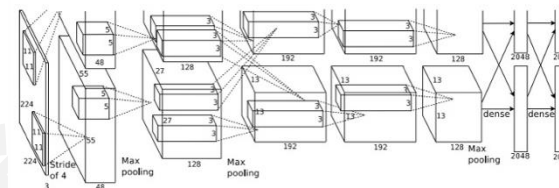
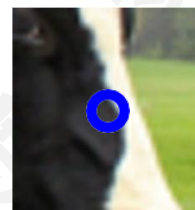
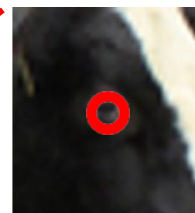
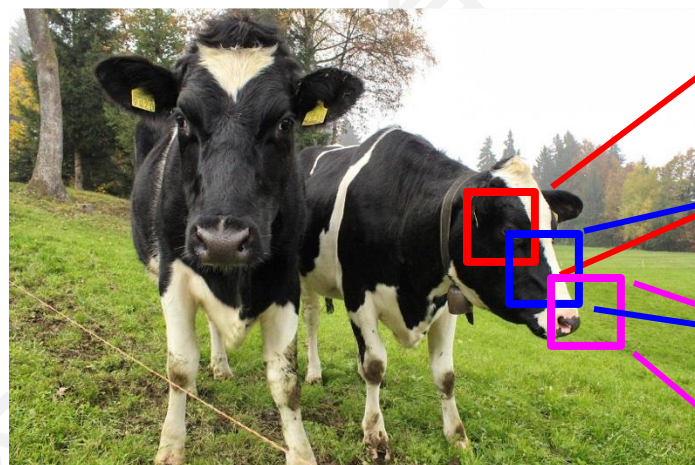


**解决方案：感知像素周围的语义，
帮助正确分类像素**

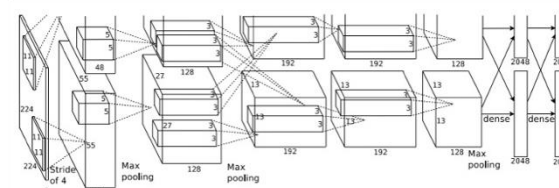
■ 语义分割-滑窗计算

图像块

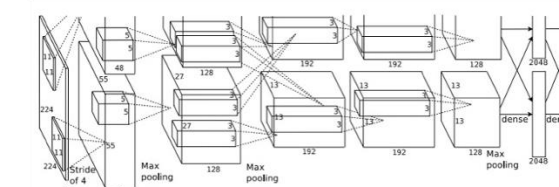
CNN



Cow



Cow

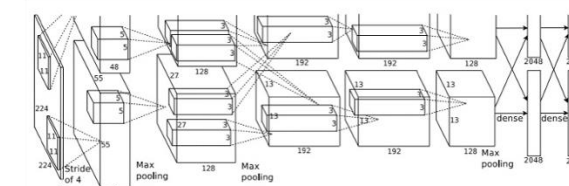
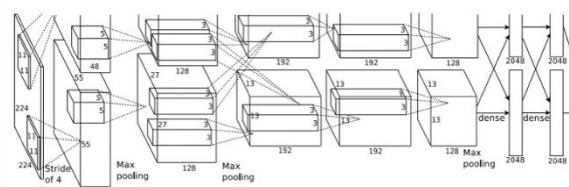
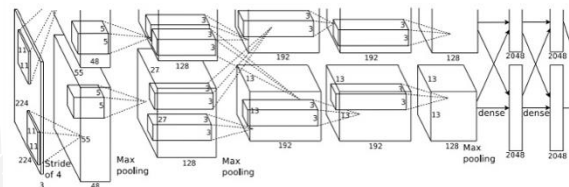
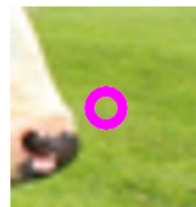
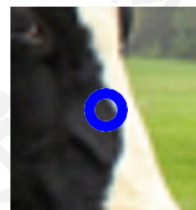
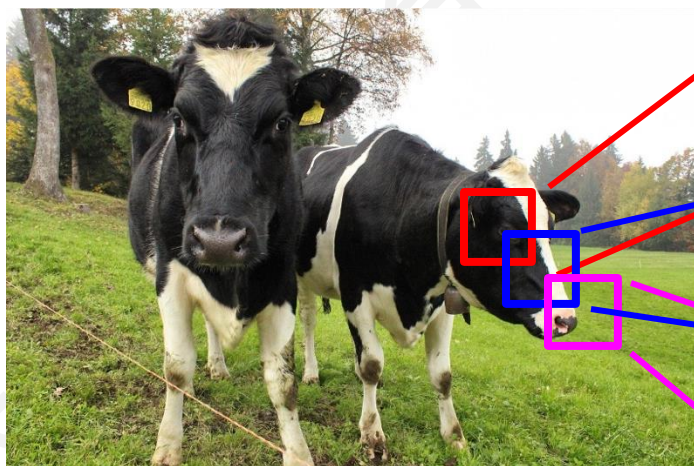


Grass

■ 语义分割-滑窗计算

图像块

CNN



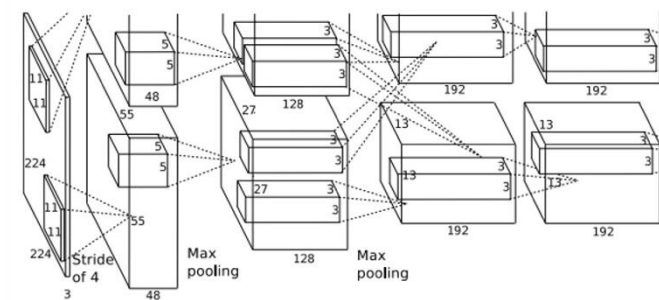
Cow

Cow

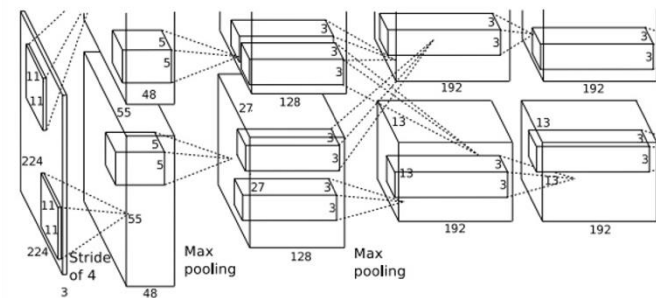
Grass

问题：计算非常低效！
图像块的重叠部分会被重复计算很多次！

■ 语义分割-全卷积

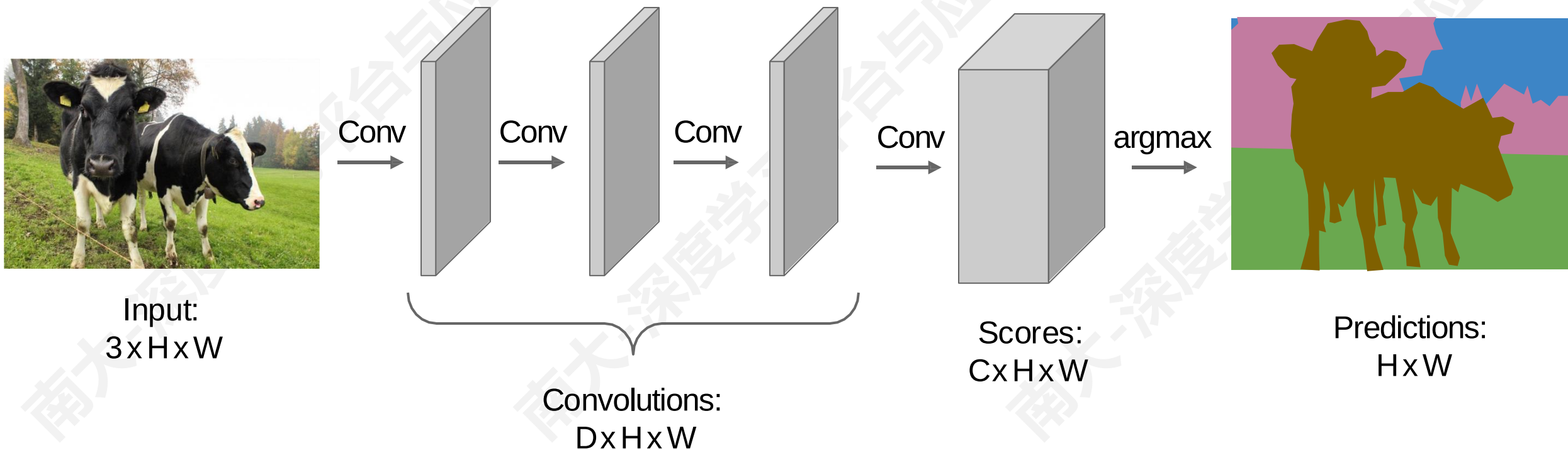


■ 语义分割-全卷积 (Fully Convolutional Networks)



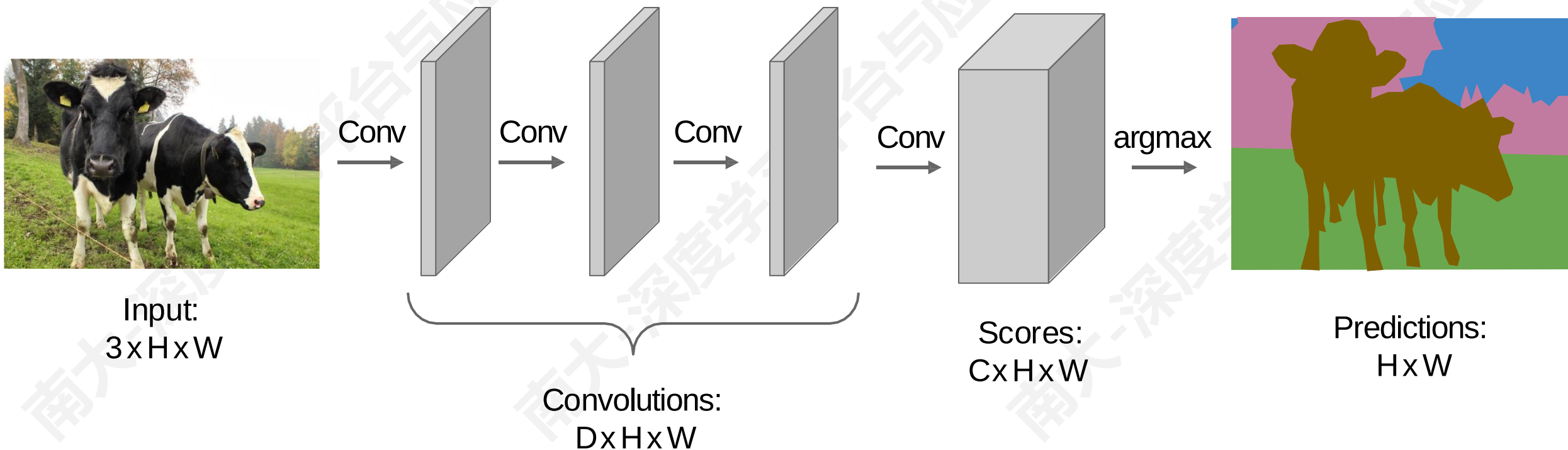
问题：分类模型会大幅降低特征的分辨率，难以满足分割所需的高分辨率输出

■ 语义分割-全卷积



解决方案：没有下采样操作的全卷积网络，维持图像高分辨率

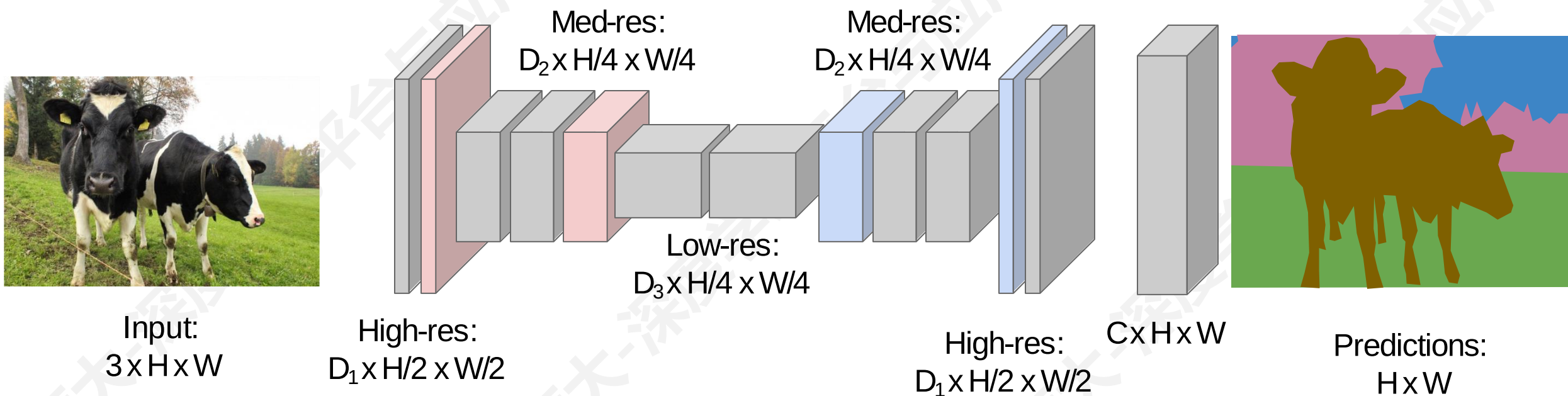
■ 语义分割-全卷积



解决方案：没有下采样操作的全卷积网络，维持图像高分辨率

问题：原图尺寸上的卷积操作的计算量/显存占用非常大

■ 语义分割-全卷积

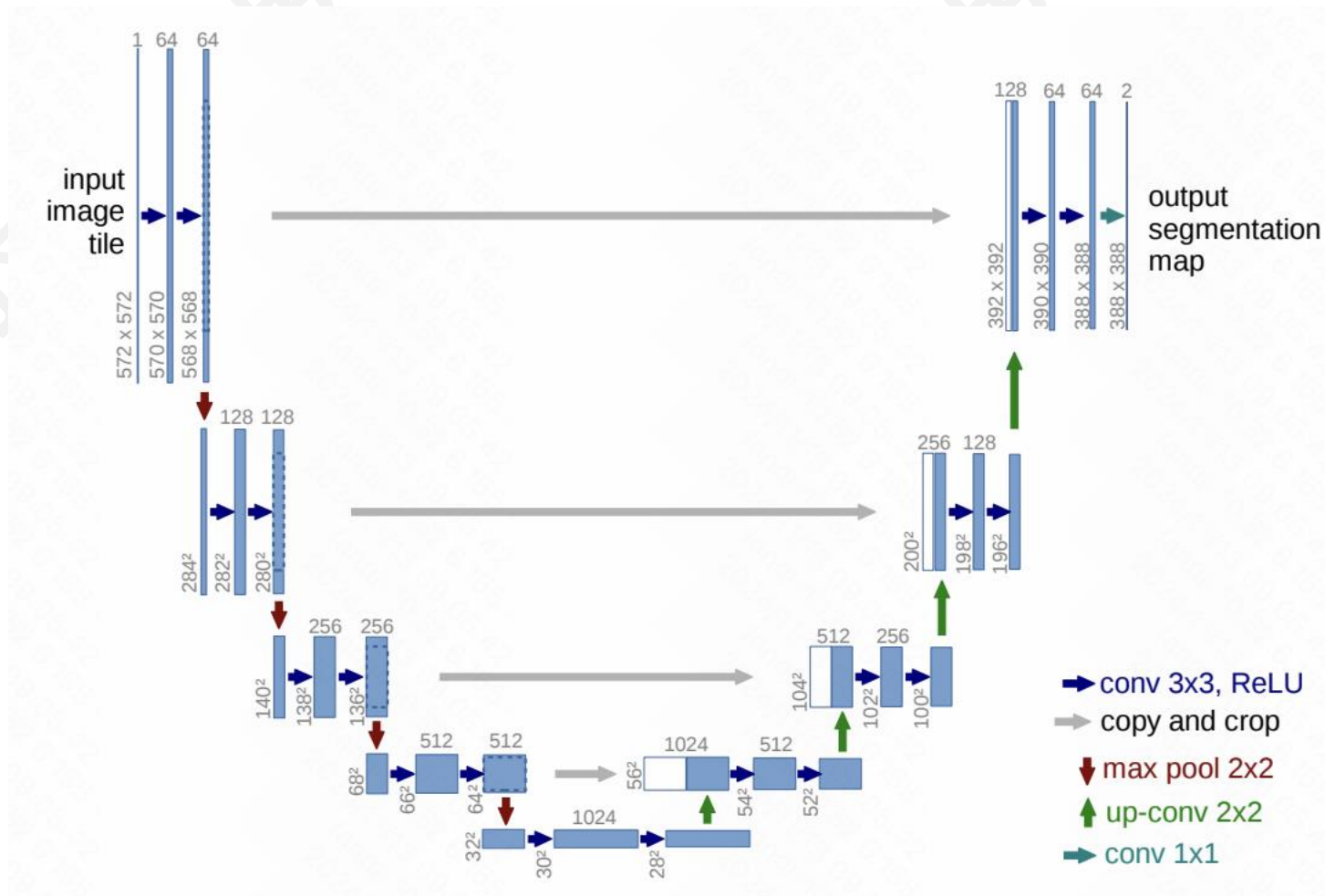


解决方案：先逐步**下采样**降低计算量，然后逐步**上采样**恢复图像分辨率

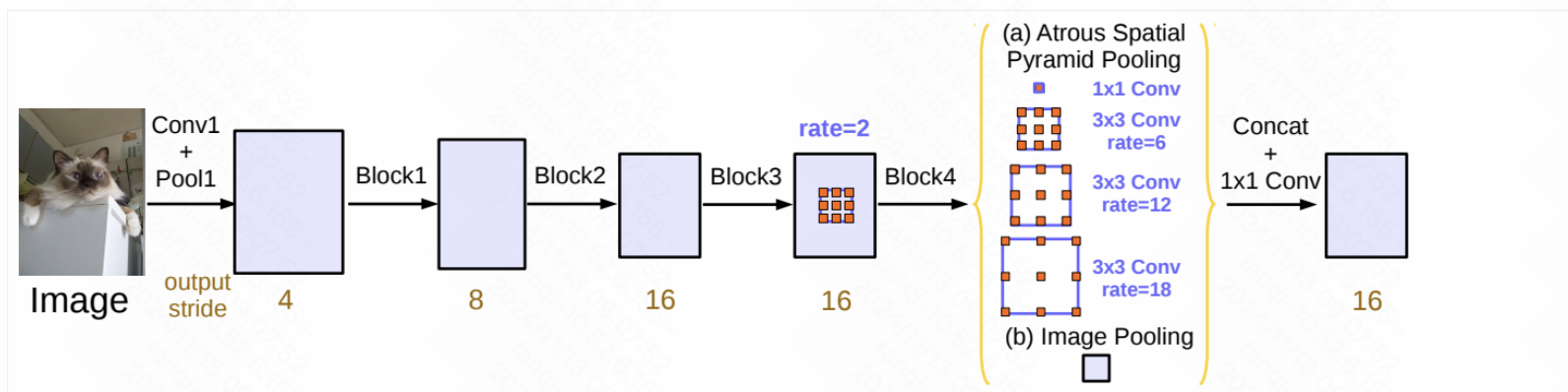
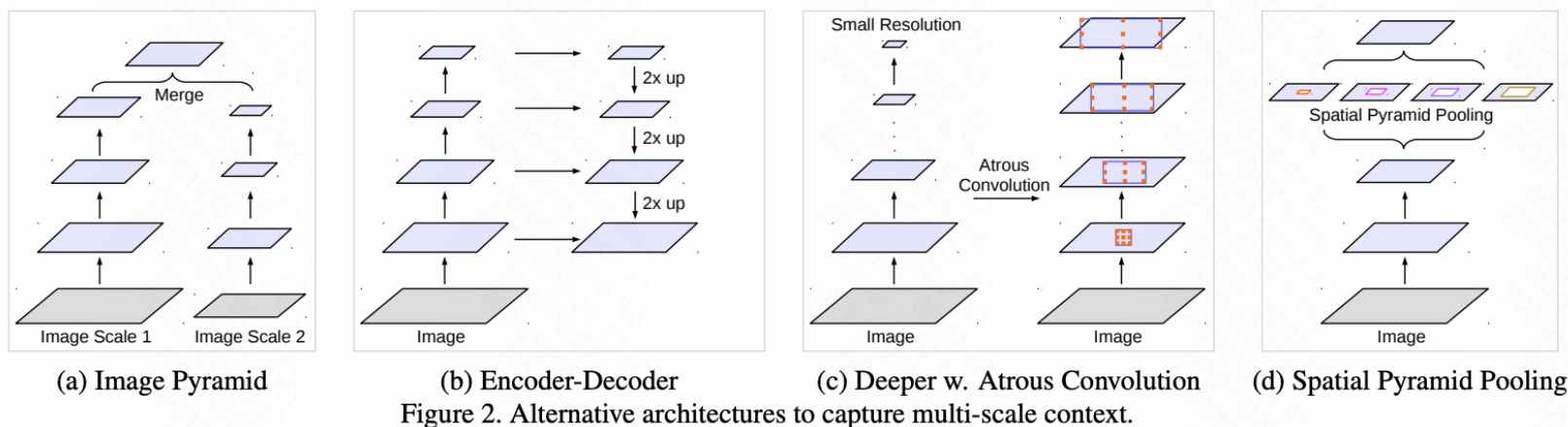
大步长的池化、卷积

Unpooling、转置卷积

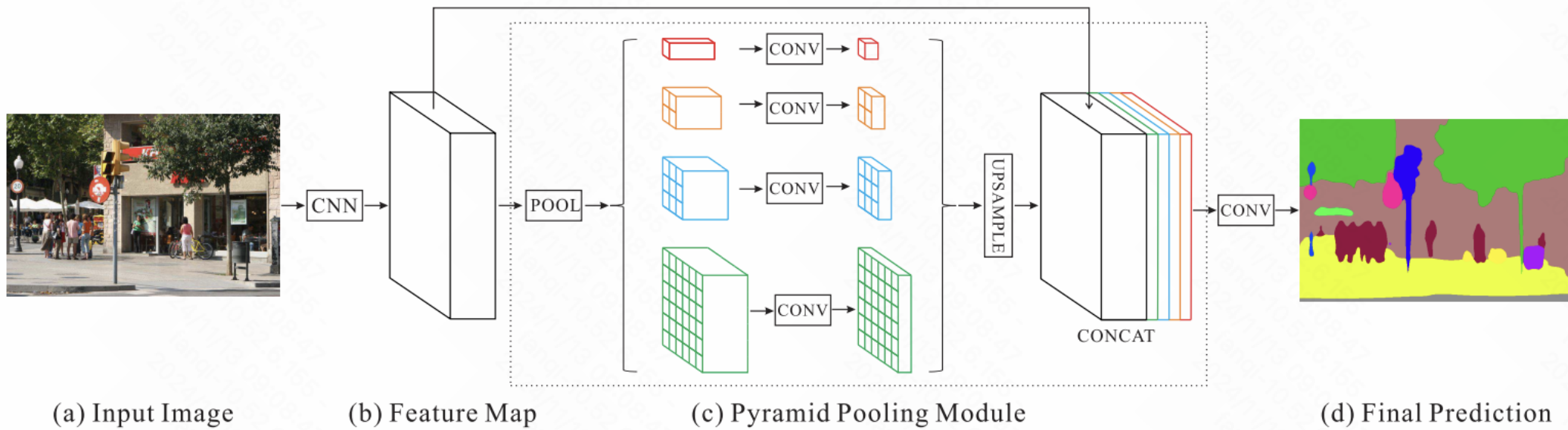
■ U-Net (Encoder-Decoder)



DeepLab (Encoder-only)

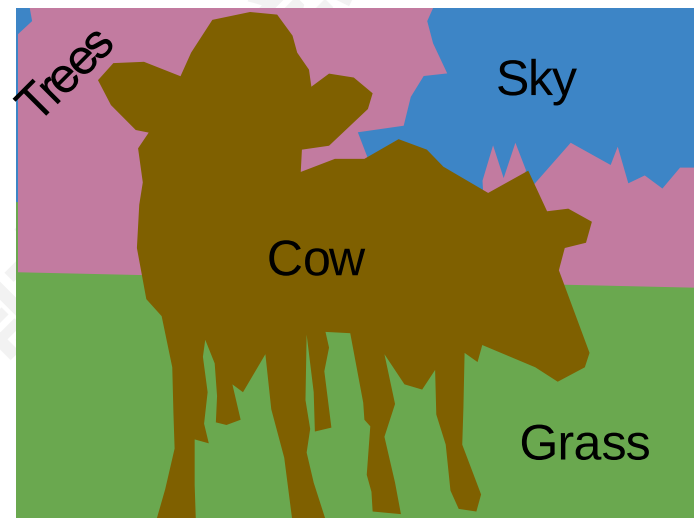
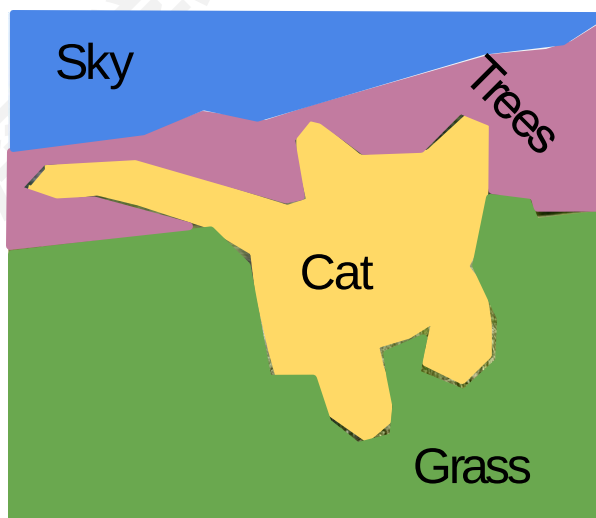


■ PSPNet (Encoder-only)



■ 语义分割-全卷积

对图像的每个像素进行分类
不区分不同物体，只关心像素



大 纲

- 语义分割
- 目标检测
- 实例分割

■ 目标检测

Classification



CAT

No spatial extent

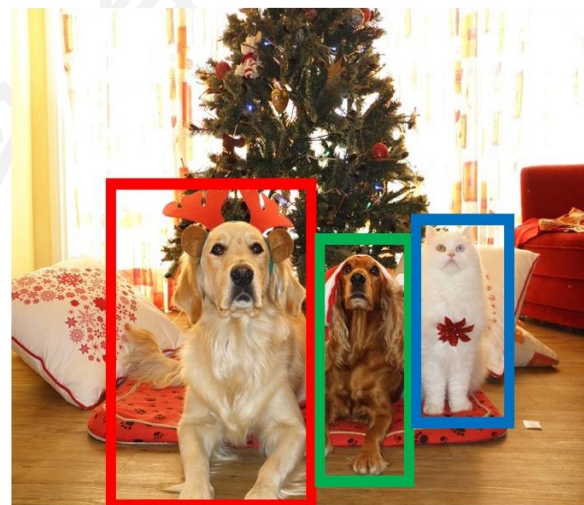
Semantic
Segmentation



GRASS, CAT, TREE,
SKY

No objects, just pixels

Object
Detection



DOG, DOG, CAT

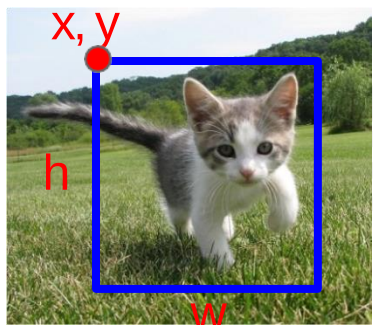
Instance
Segmentation



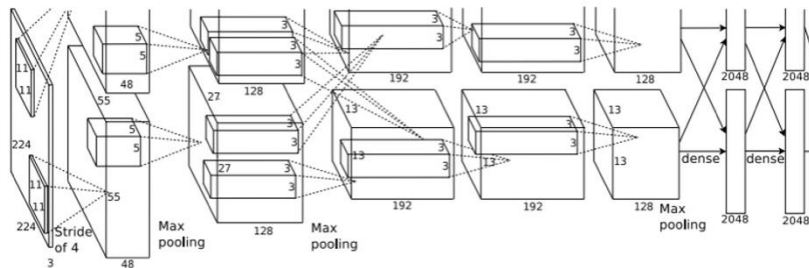
DOG, DOG, CAT

Multiple Object

■ 目标检测-单个物体：分类+定位



[This image is CC0 public domain](#)



Fully
Connected:
4096 to 1000

Class Scores

Cat: 0.9

Dog: 0.05

Car: 0.01

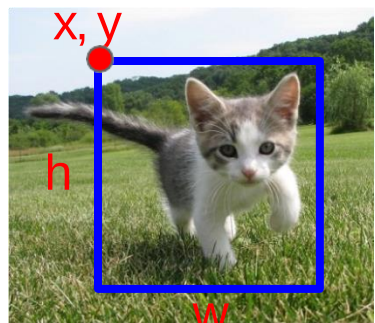
...

Vector:
4096

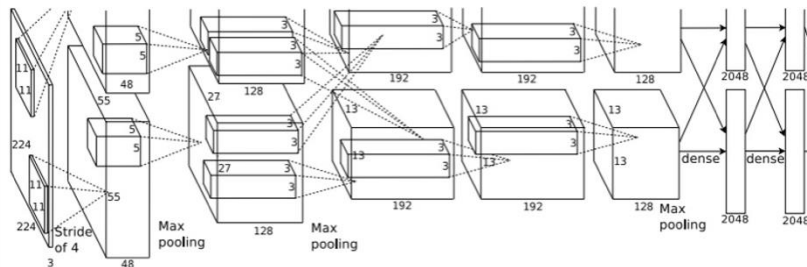
Fully
Connected:
4096 to 4

Box
Coordinates
(x, y, w, h)

目标检测-单个物体：分类+定位



[This image is CC0 public domain](#)



Fully
Connected:
4096 to 1000

Class Scores

Cat: 0.9
Dog: 0.05
Car: 0.01
...

Correct label:
Cat

Softmax
Loss

Vector:
4096

Fully
Connected:
4096 to 4

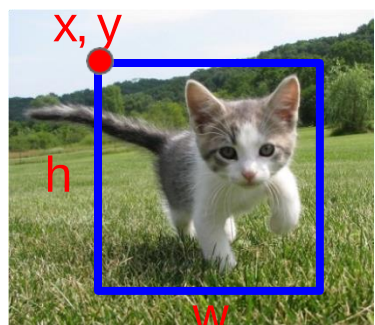
Box
Coordinates
(x, y, w, h)

L2 Loss

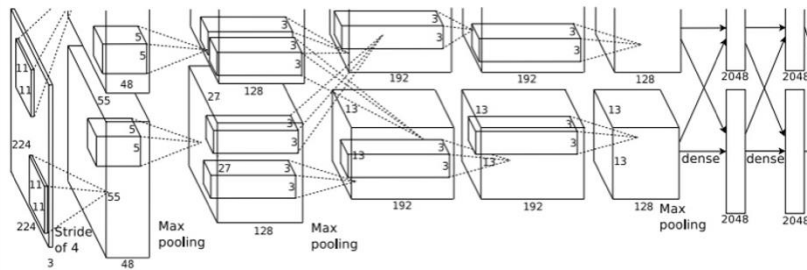
Correct box:
(x', y', w', h')

将定位建模为回归任务

■ 目标检测-单个物体：分类+定位



[This image is CC0 public domain](#)



Fully Connected:
4096 to 1000

Class Scores

Cat: 0.9
Dog: 0.05
Car: 0.01
...

Correct label:
Cat

Softmax
Loss

Multitask Loss

+ → Loss

Vector:
4096

Fully Connected:
4096 to 4

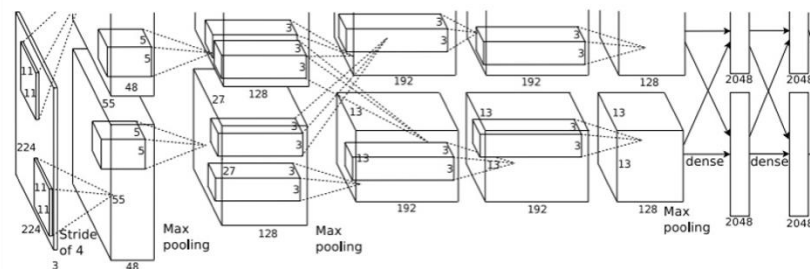
Box
Coordinates
(x, y, w, h)

L2 Loss

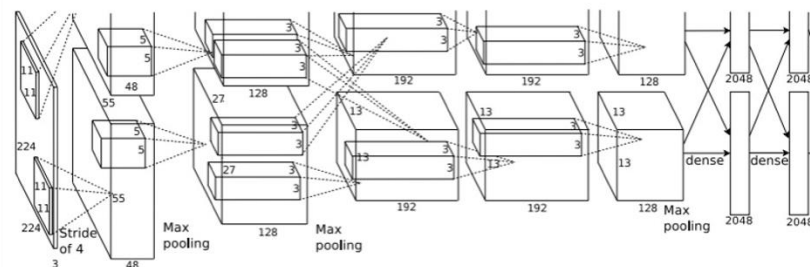
Correct box:
(x', y', w', h')

将定位建模为回归任务

目标检测-多个物体



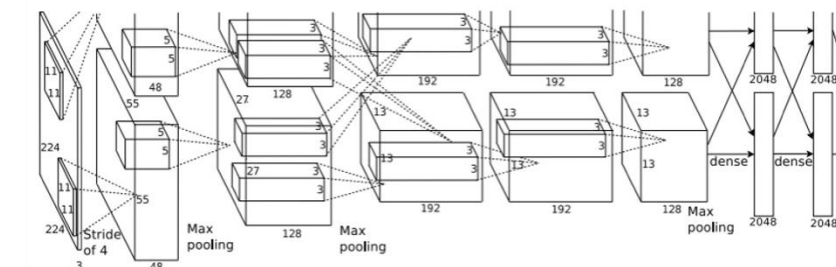
CAT: (x, y, w, h)



DOG: (x, y, w, h)

DOG: (x, y, w, h)

CAT: (x, y, w, h)



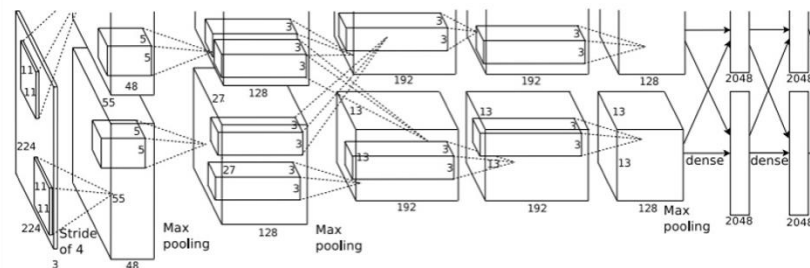
DUCK: (x, y, w, h)

DUCK: (x, y, w, h)

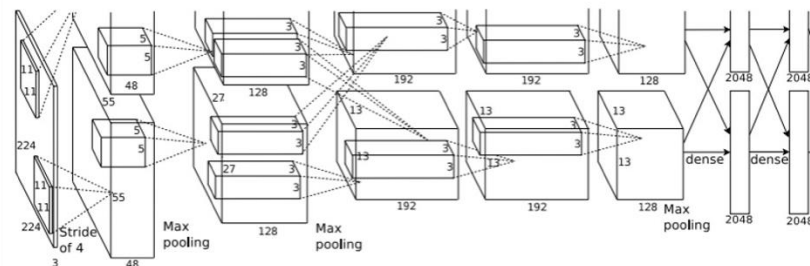
....

目标检测-多个物体

问题：每个图像都有不同数量的输出



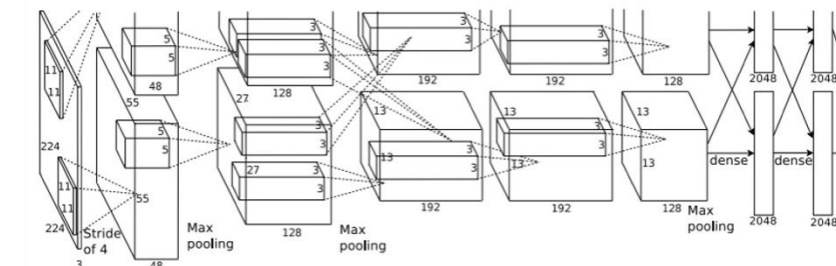
CAT: (x, y, w, h)



DOG: (x, y, w, h)

DOG: (x, y, w, h)

CAT: (x, y, w, h)



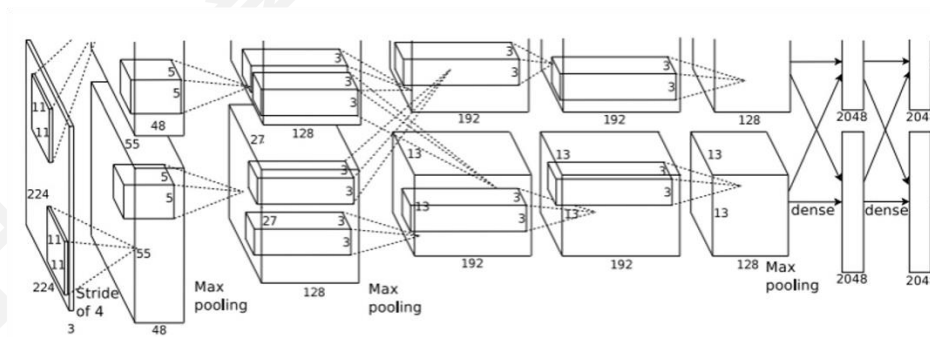
DUCK: (x, y, w, h)

DUCK: (x, y, w, h)

....

■ 目标检测-多个物体

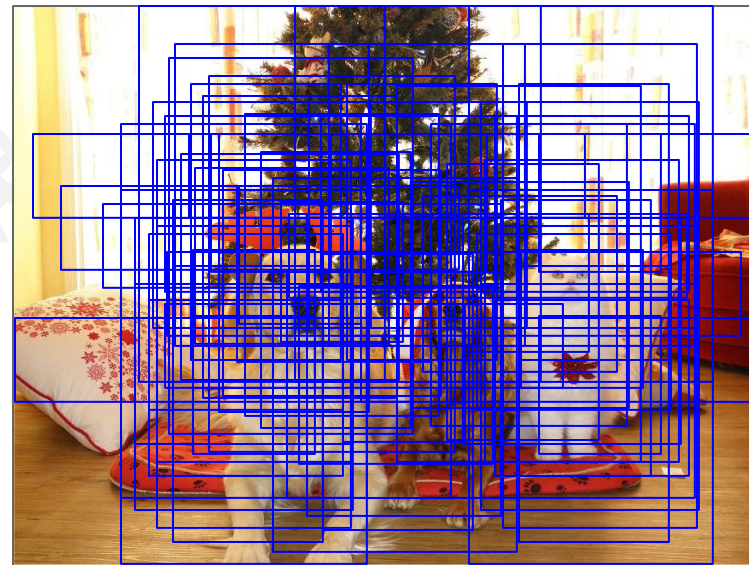
将图像分为不同图像区域，对每个图像区域进行单个物体检测



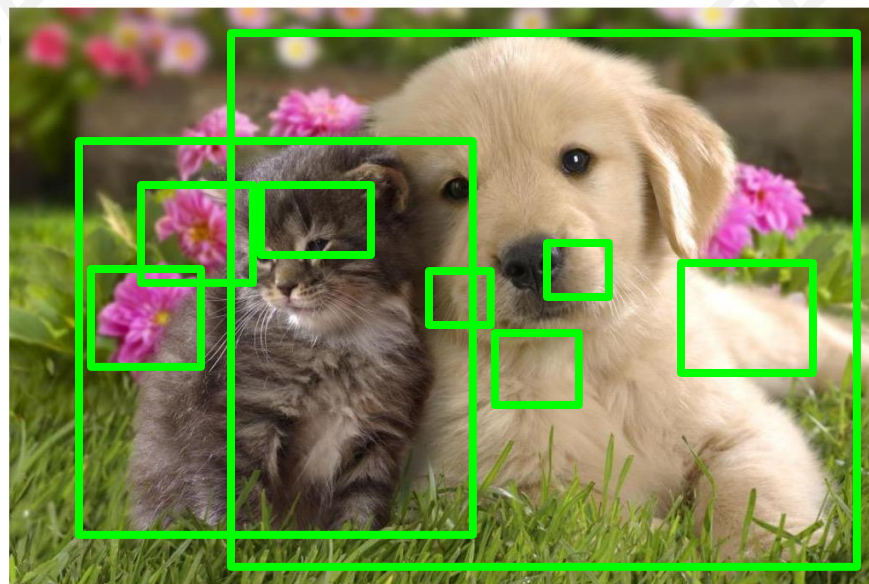
Dog? NO
Cat? NO
Background? YES

■ 目标检测-多个物体

问题：如何快速确定图像区域？
图像区域有不同的位置、大小、比例



■ Region Proposals: Selective Search



找到有可能存在物体的区域

Selective Search 在 CPU 上运行，可以在几秒内生成 2000 多个区域

■ R-CNN

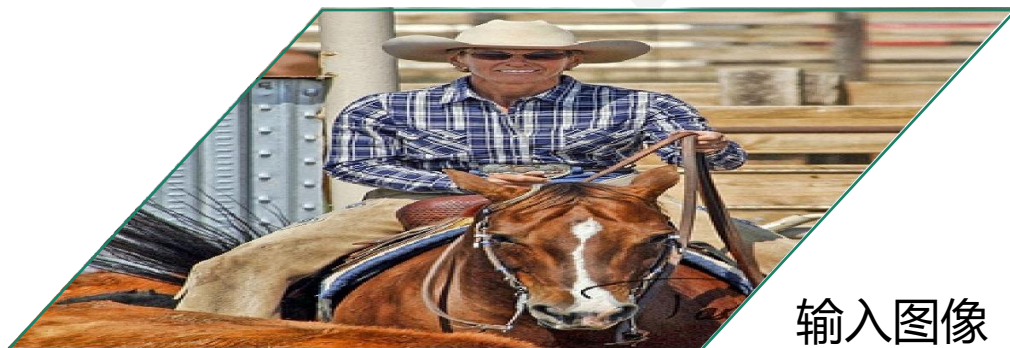


arXiv

<https://arxiv.org> › cs · [翻译此页](#) ⋮

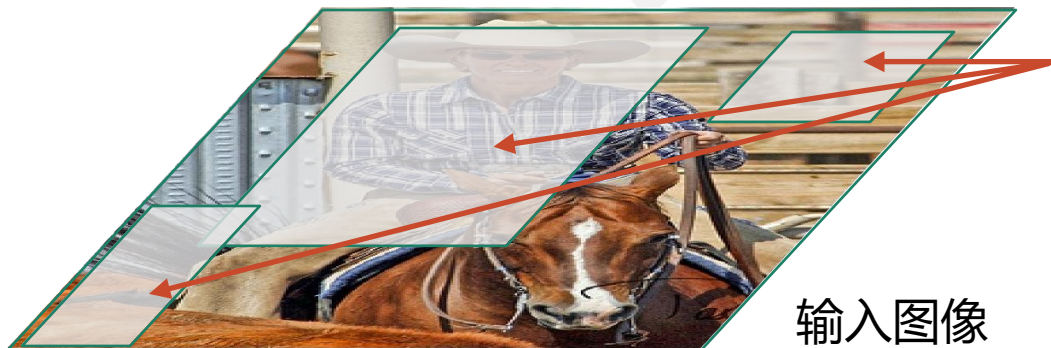
Rich feature hierarchies for accurate object detection and ...

作者: R Girshick · 2013 · 被引用次数: 39820 — Since we combine region proposals with CNNs, we call our method R-CNN: Regions with CNN features. We also compare R-CNN to...



输入图像

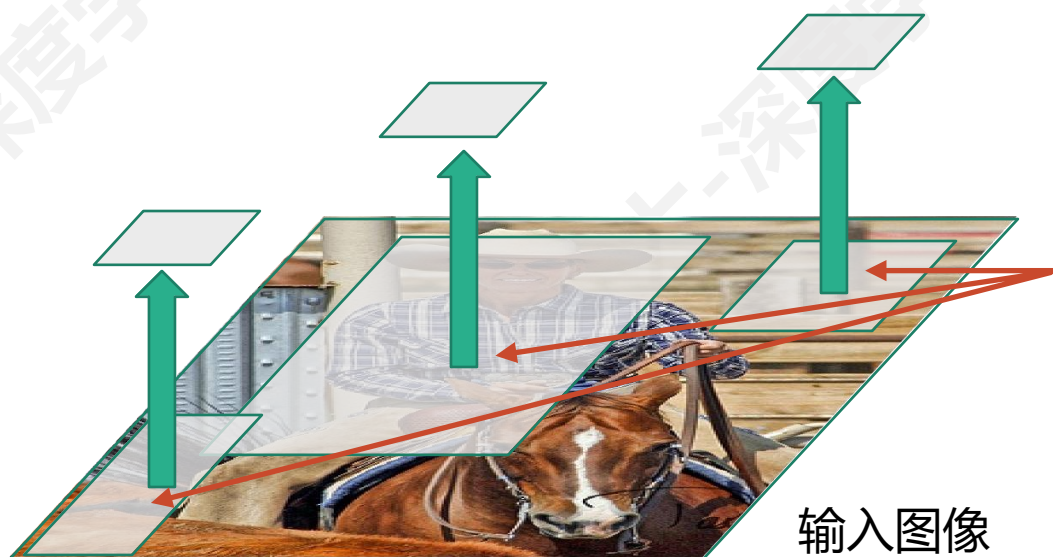
■ R-CNN



输入图像

由区域生成网络得到
Regions of Interest (RoI)
(~2k)

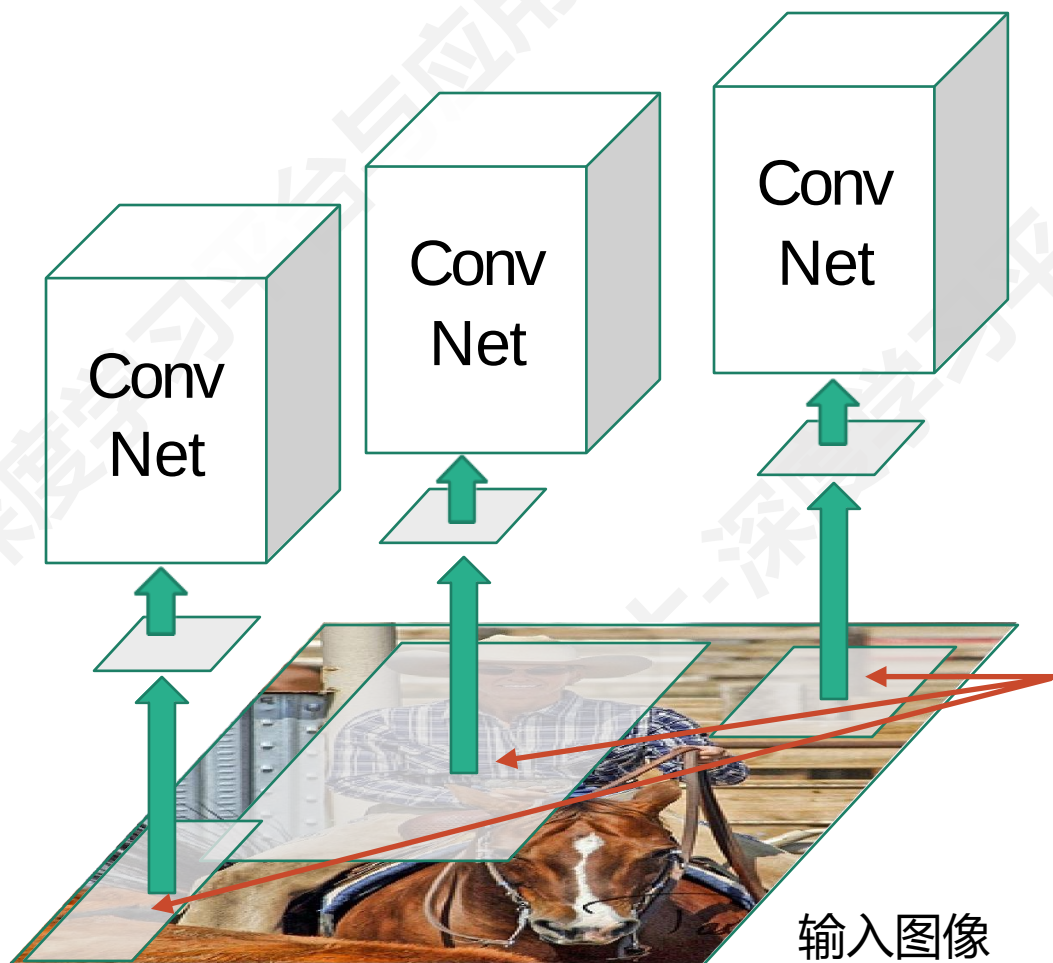
■ R-CNN



Wrap 处理后的图像区域

由区域生成网络得到
Regions of Interest (RoI)
(~2k), proposal

■ R-CNN

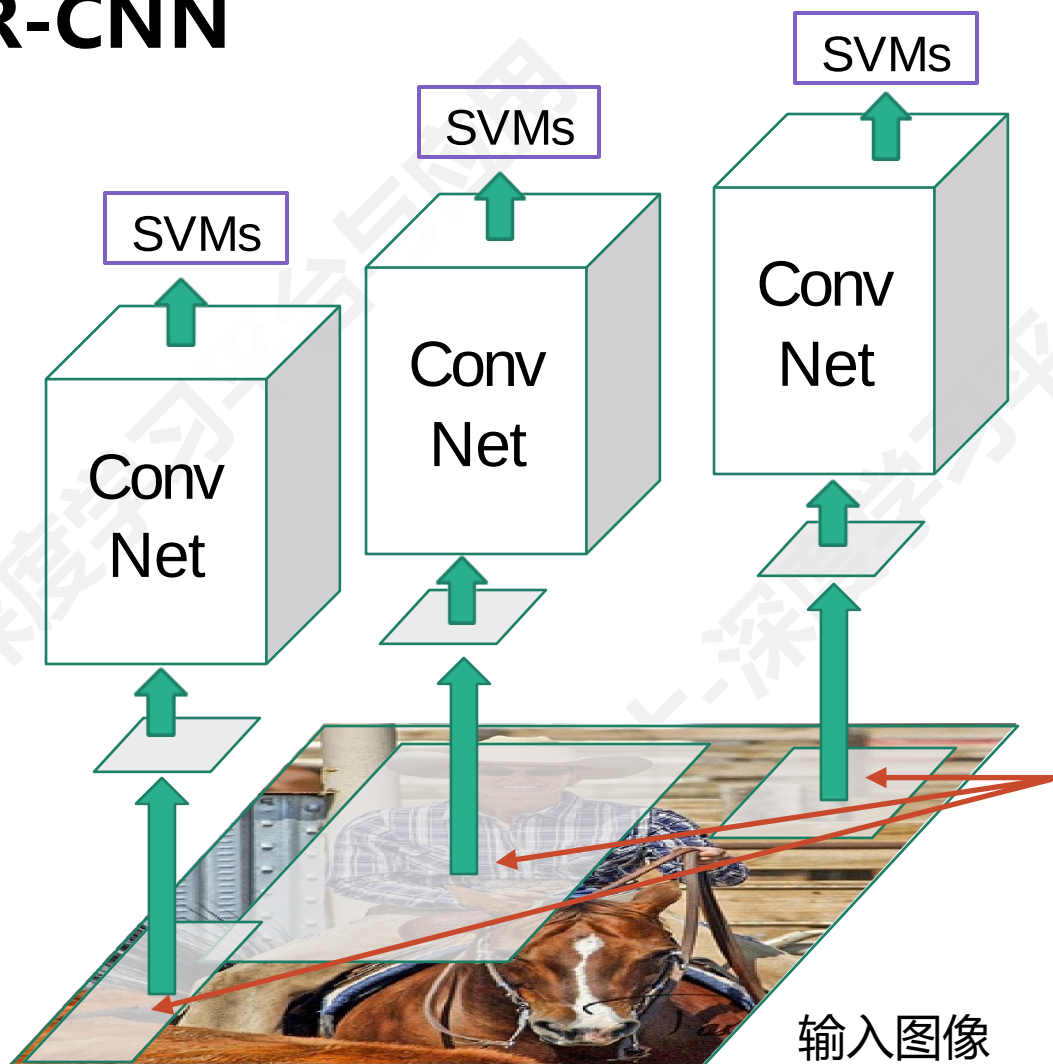


每个图像区域使用卷积网络处理
ImageNet 预训练的 CNN

Wrap 处理后的图像区域

由区域生成网络得到
Regions of Interest (RoI)
(~2k), proposal

■ R-CNN



使用SVM对每个图像区域分类

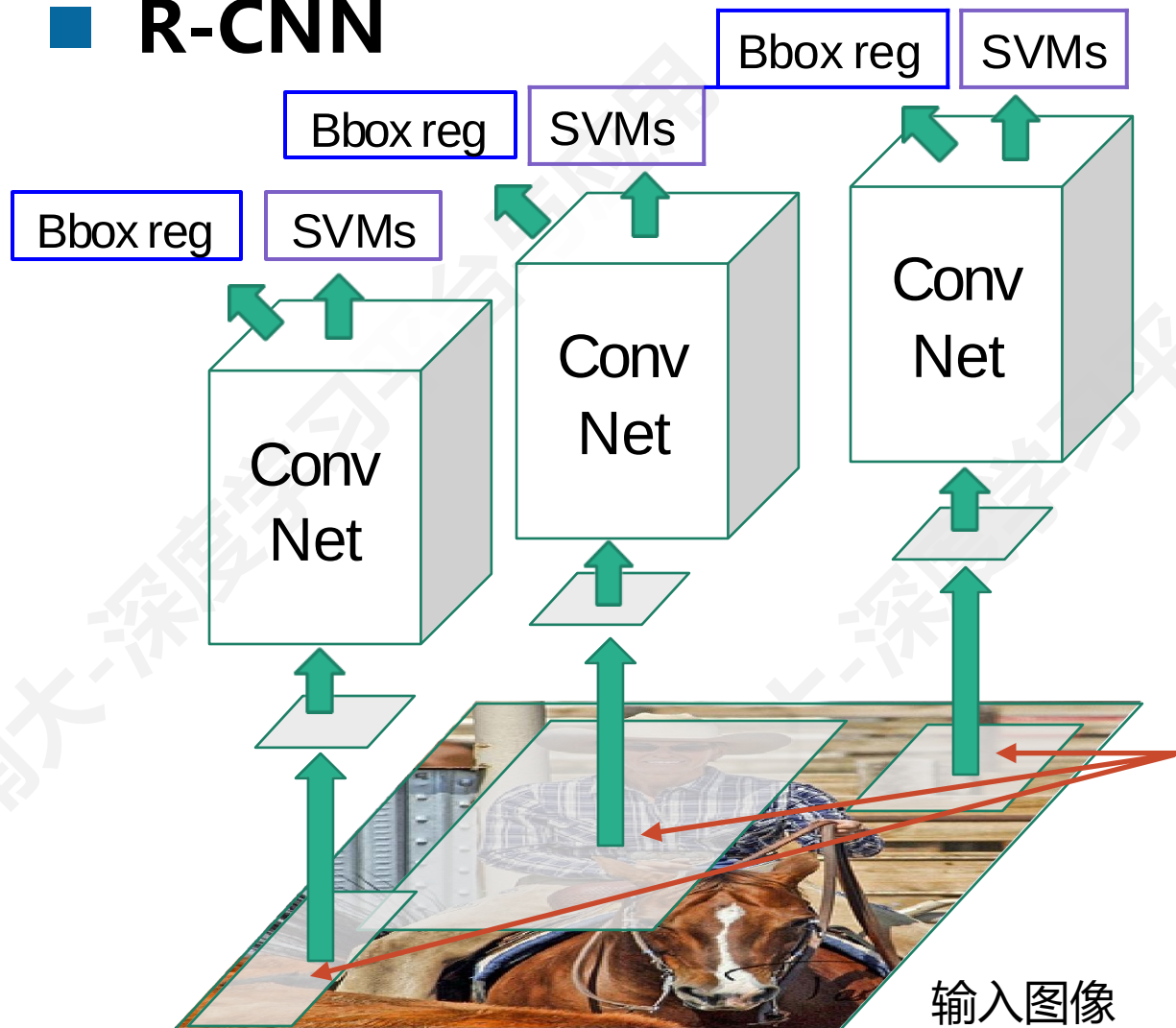
每个图像区域使用卷积网络处理
ImageNet 预训练的 CNN

Wrap 处理后的图像区域

由区域生成网络得到
Regions of Interest (RoI)
(~2k), proposal

对每个 RoI 预测调整值: (dx, dy, dw, dh)

■ R-CNN



使用SVM对每个图像区域分类

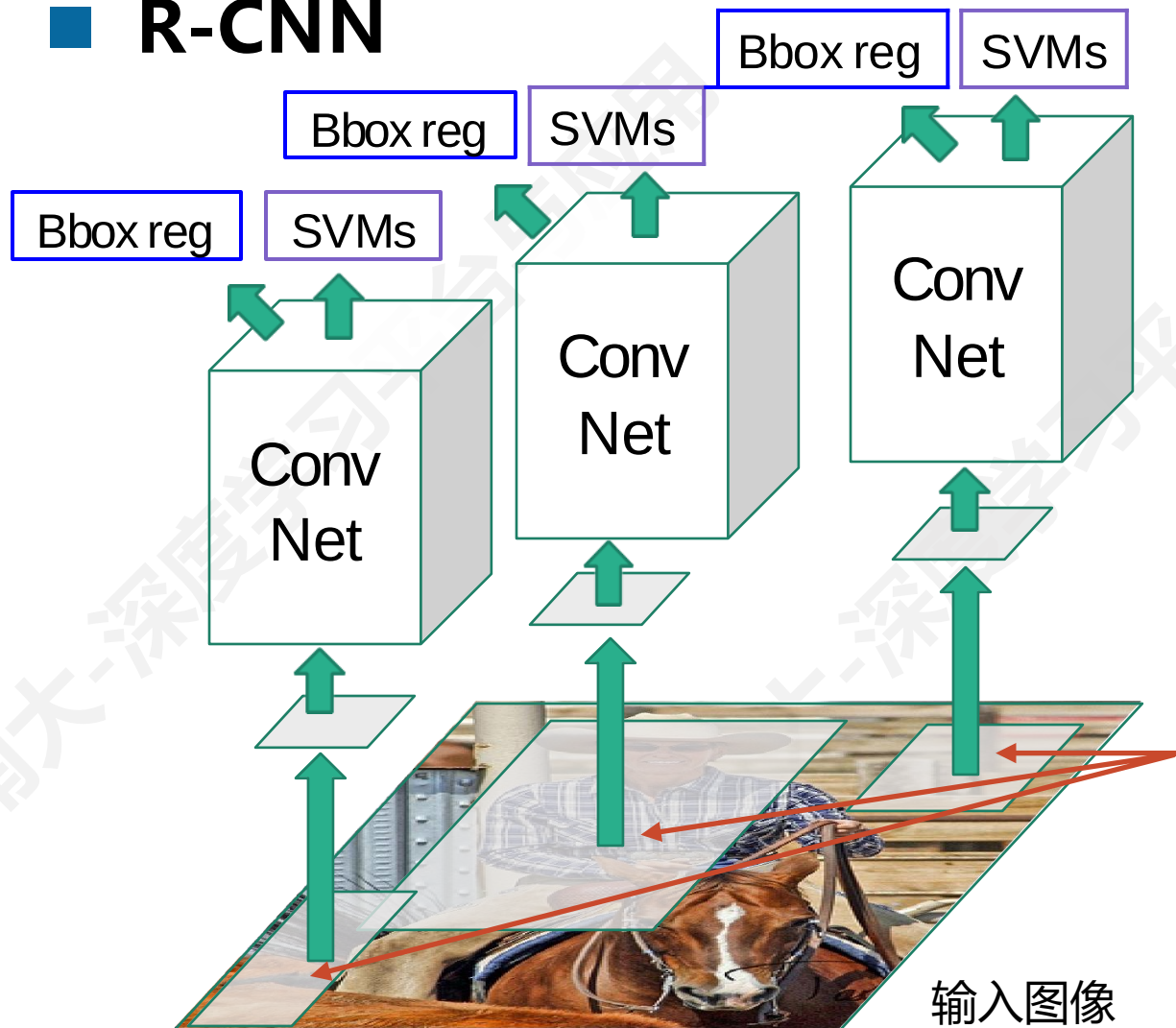
每个图像区域使用卷积网络处理
ImageNet 预训练的 CNN

Wrap 处理后的图像区域

由区域生成网络得到
Regions of Interest (RoI)
(~2k), proposal

对每个 RoI 预测调整值: (dx, dy, dw, dh)

■ R-CNN



使用SVM对每个图像区域分类

**问题：效率很低！非常慢！
同一块区域会被反复计算**

每个图像区域使用卷积网络处理
ImageNet 预训练的 CNN

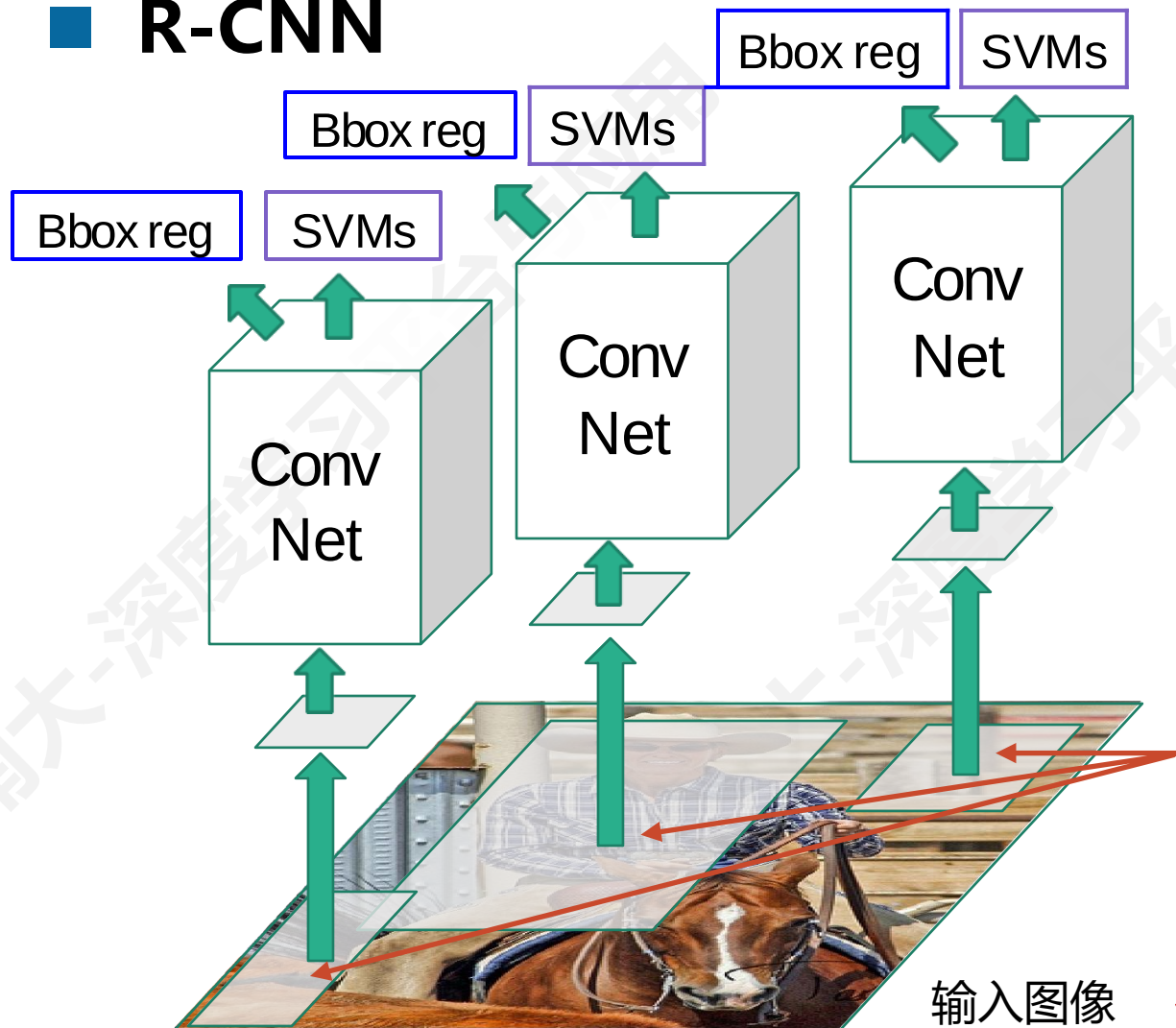
Wrap 处理后的图像区域

由区域生成网络得到
Regions of Interest (RoI)
(~2k), proposal

输入图像

对每个 RoI 预测调整值: (dx, dy, dw, dh)

■ R-CNN



使用SVM对每个图像区域分类

问题：效率很低！非常慢！
同一块区域会被反复计算

每个图像区域使用卷积网络处理
ImageNet 预训练的 CNN

Wrap 处理后的图像区域

由区域生成网络得到
Regions of Interest (RoI)
(~2k), proposal

**Idea：使用卷积网络处理整张图像，
在特征上进行区域提取**

■ Fast R-CNN



arXiv

<https://arxiv.org> > cs · [翻译此页](#) ⋮

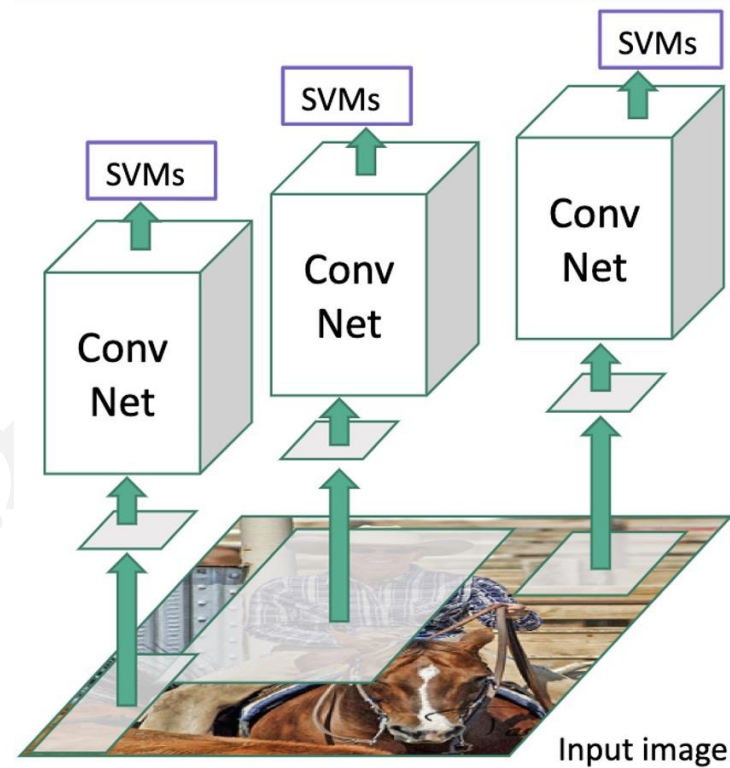
[1504.08083] Fast R-CNN

作者: R Girshick · 2015 · 被引用次数: 37400 — **Fast R-CNN** trains the very deep VGG16 network 9x faster than R-CNN, is 213x faster at test-time, and achieves a higher mAP on...

骨干网络:
AlexNet, VGG,
ResNet, ...



“Slow” R-CNN



■ Fast R-CNN

由区域生成网络得到
Regions of Interest (RoI)
(~2k), proposal

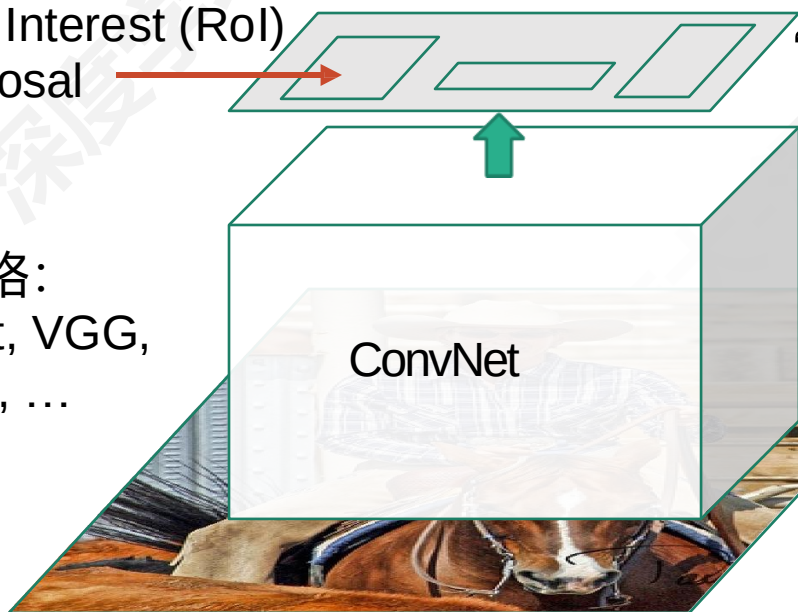
“conv5” features

使用卷积网络
处理整张图像

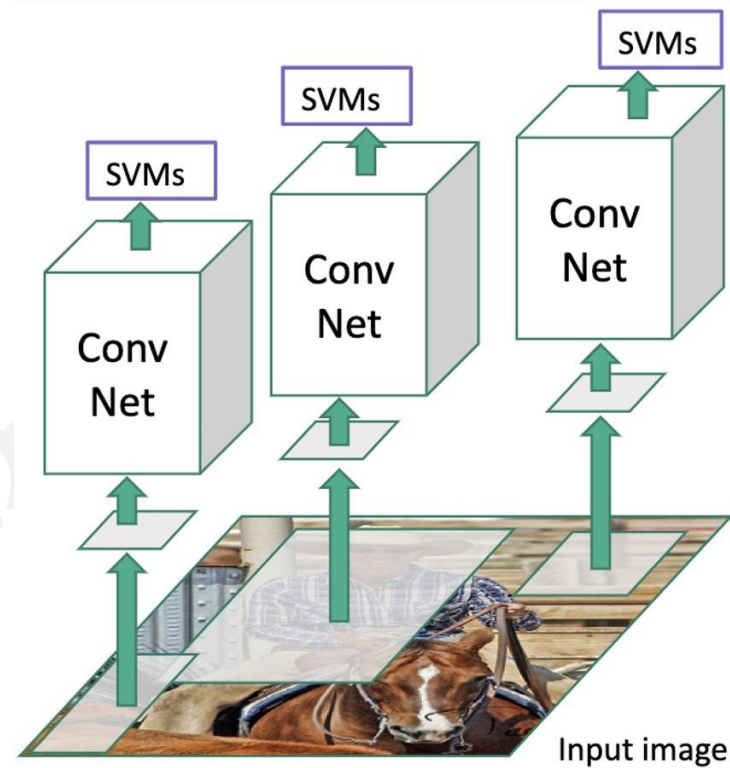
骨干网络:
AlexNet, VGG,
ResNet, ...

ConvNet

输入图像



“Slow” R-CNN



■ Fast R-CNN

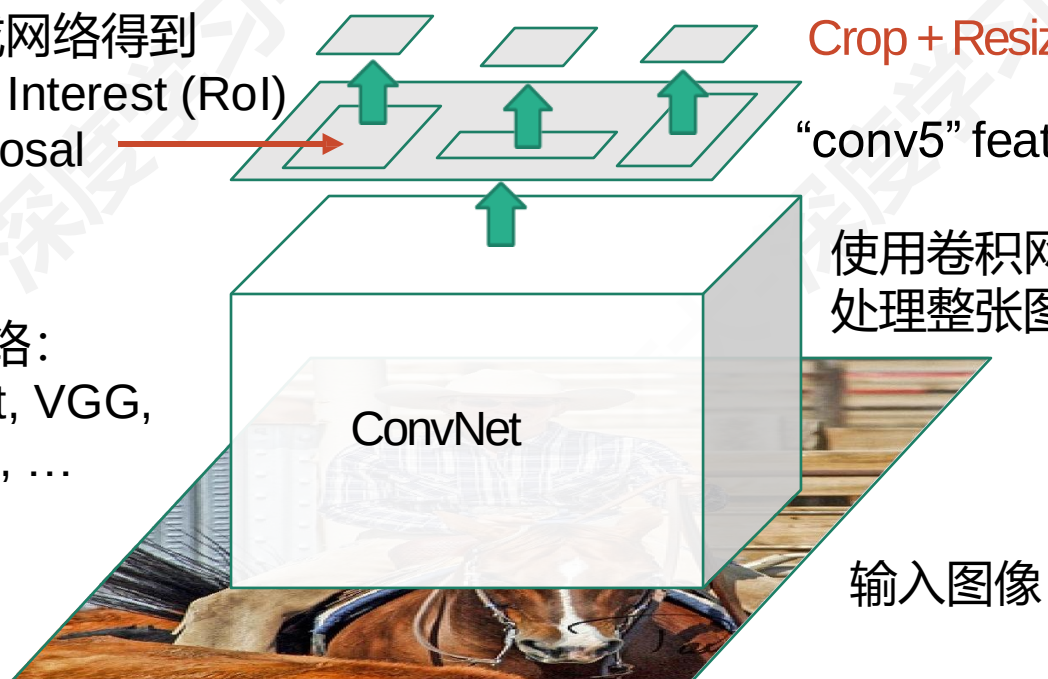
由区域生成网络得到
Regions of Interest (RoI)
(~2k), proposal

Crop + Resize features

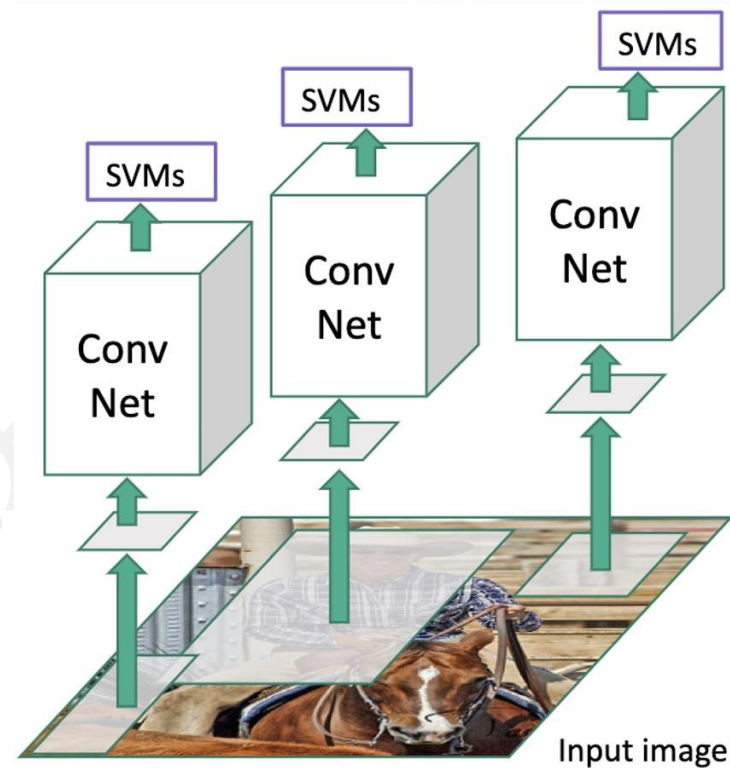
“conv5” features

使用卷积网络
处理整张图像

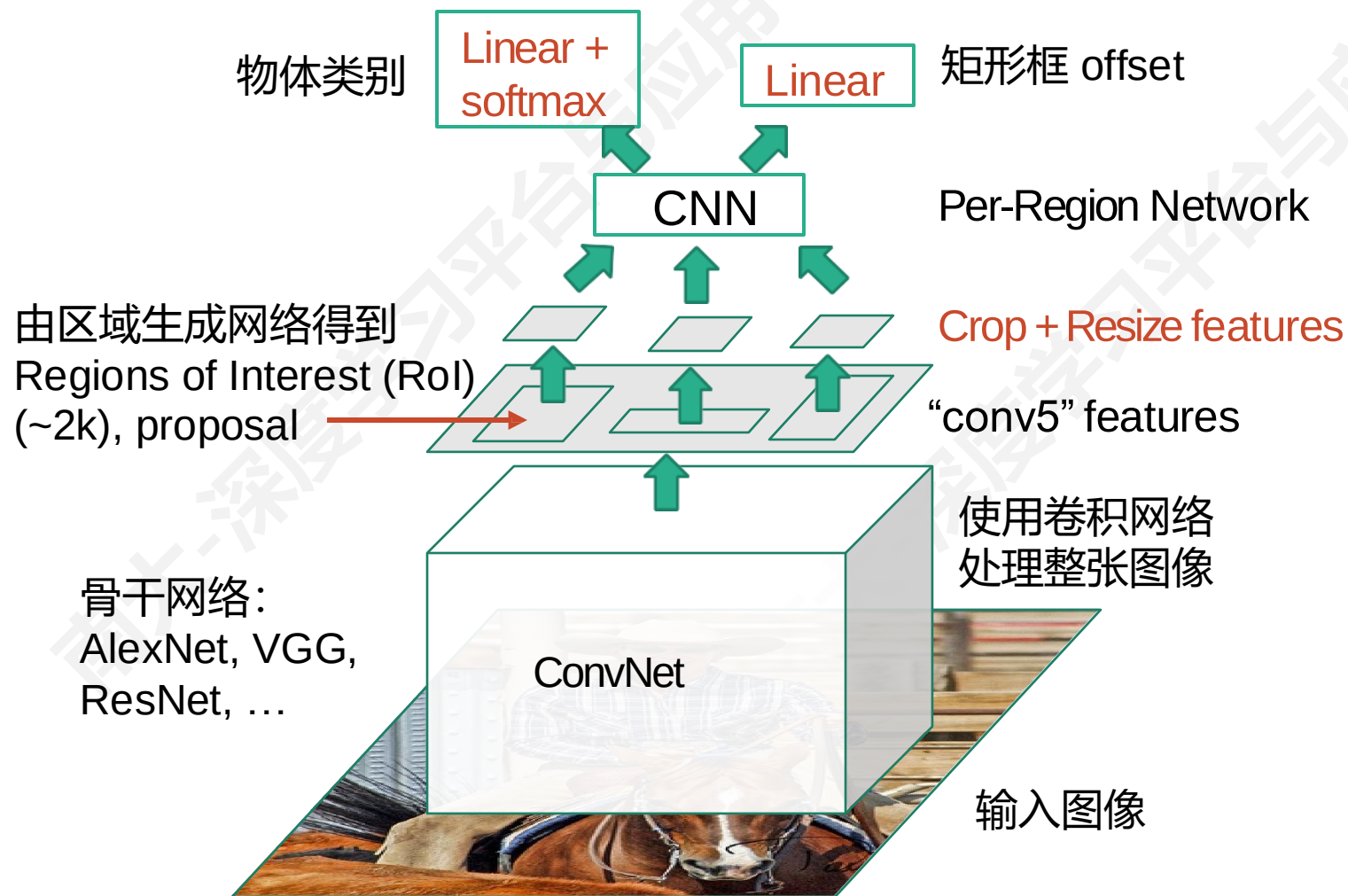
骨干网络:
AlexNet, VGG,
ResNet, ...



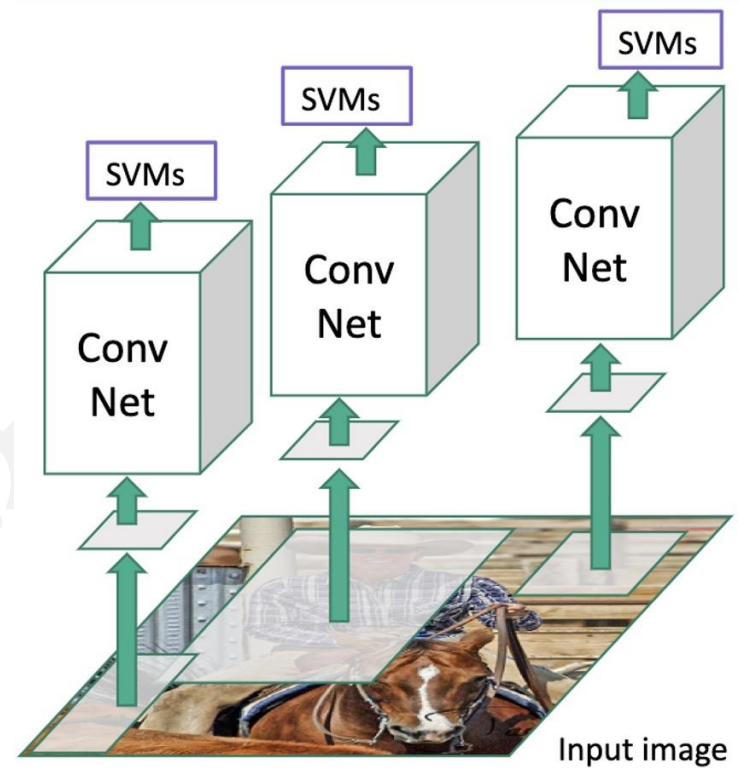
“Slow” R-CNN



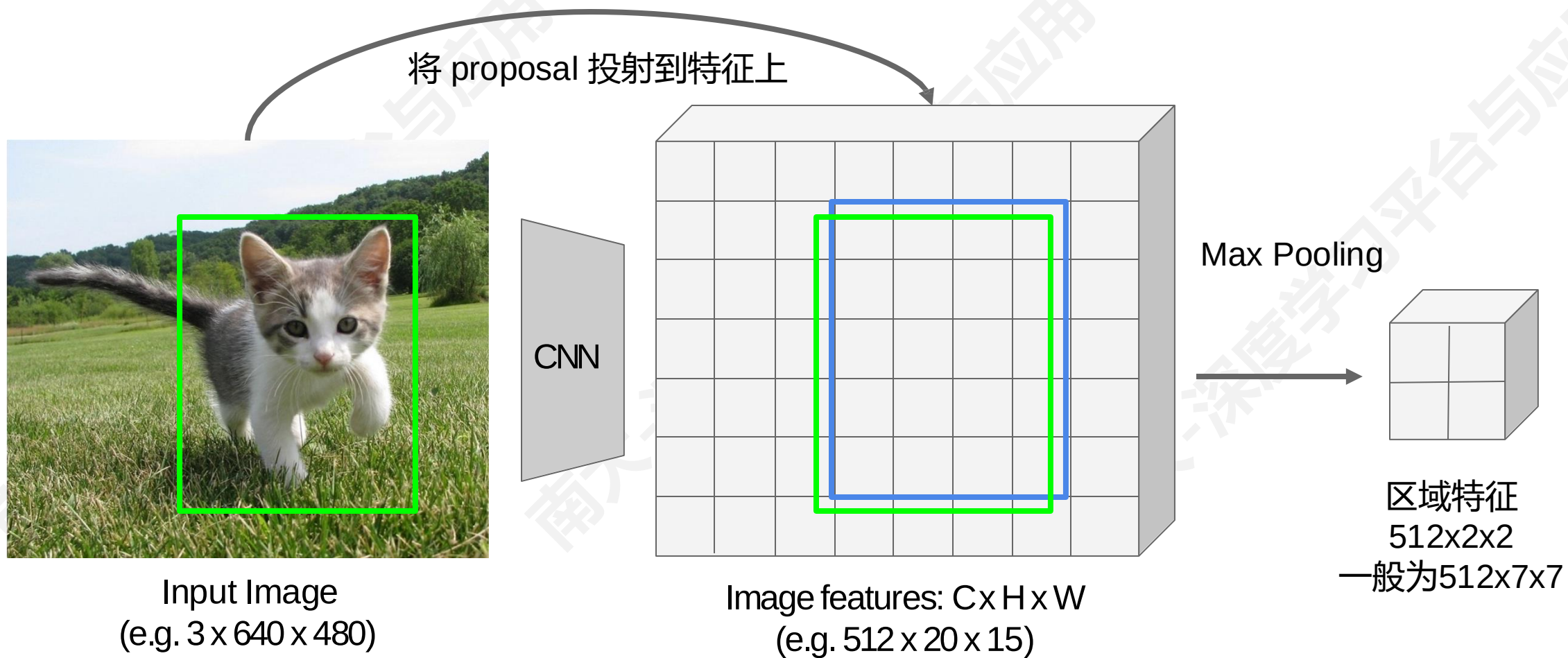
Fast R-CNN



“Slow” R-CNN

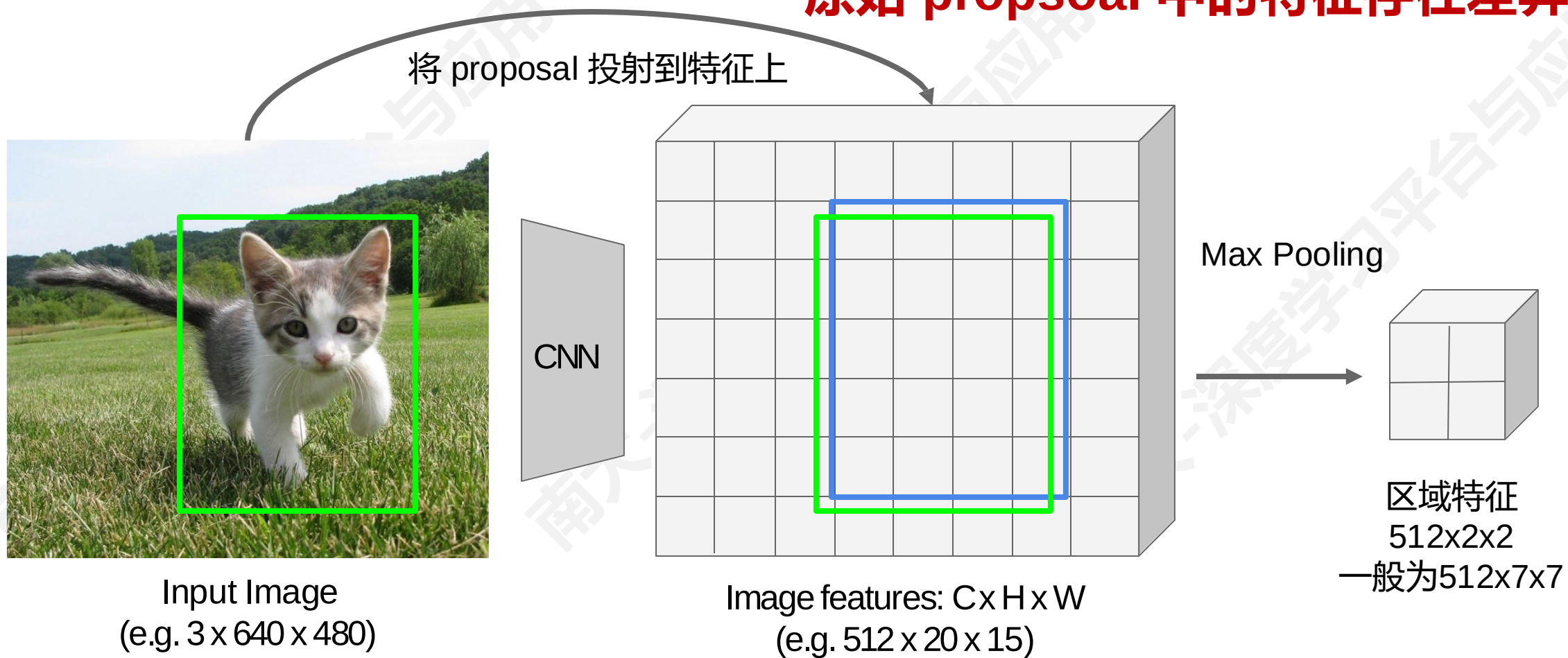


■ RoI Pooling

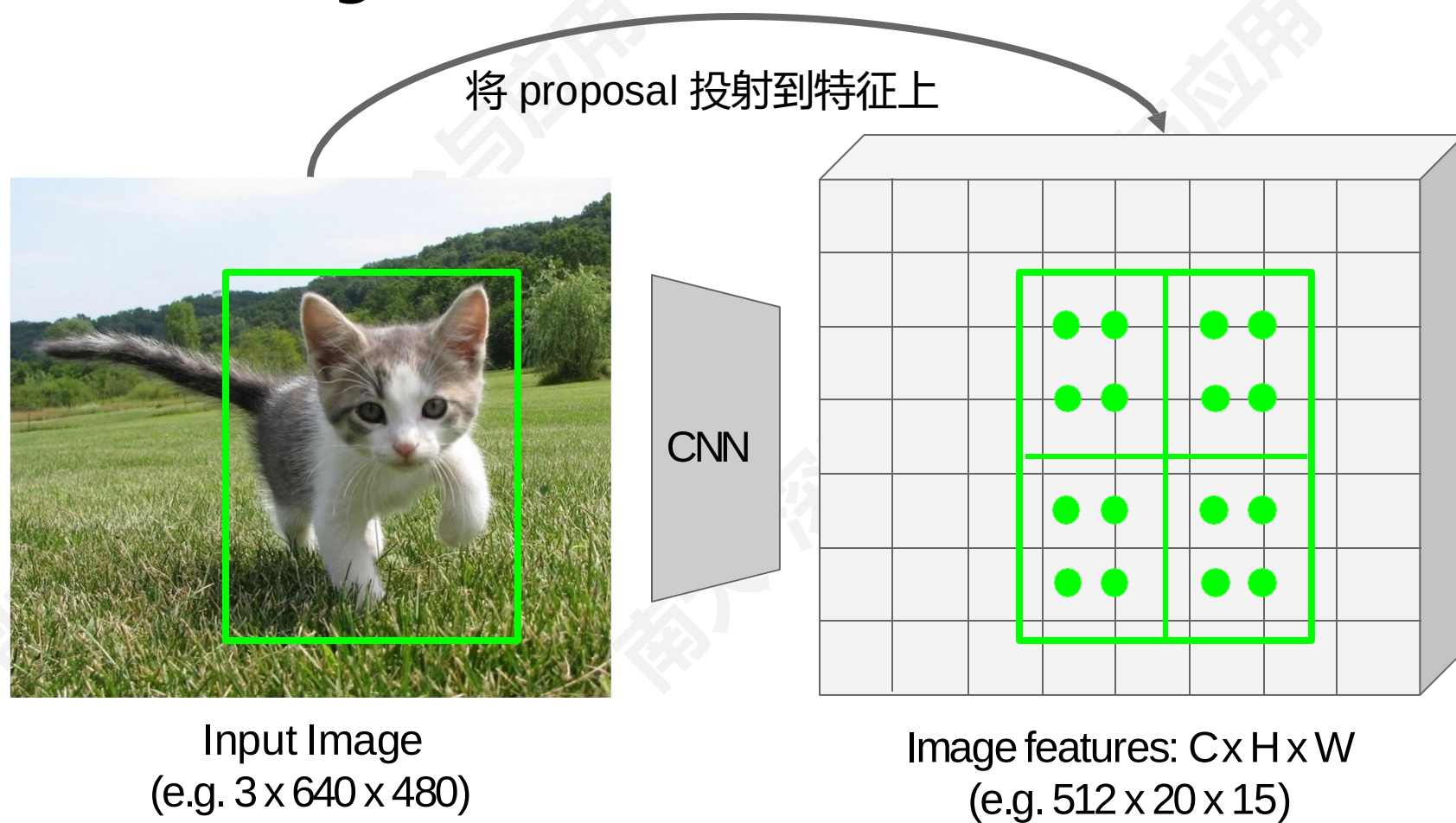


■ RoI Pooling

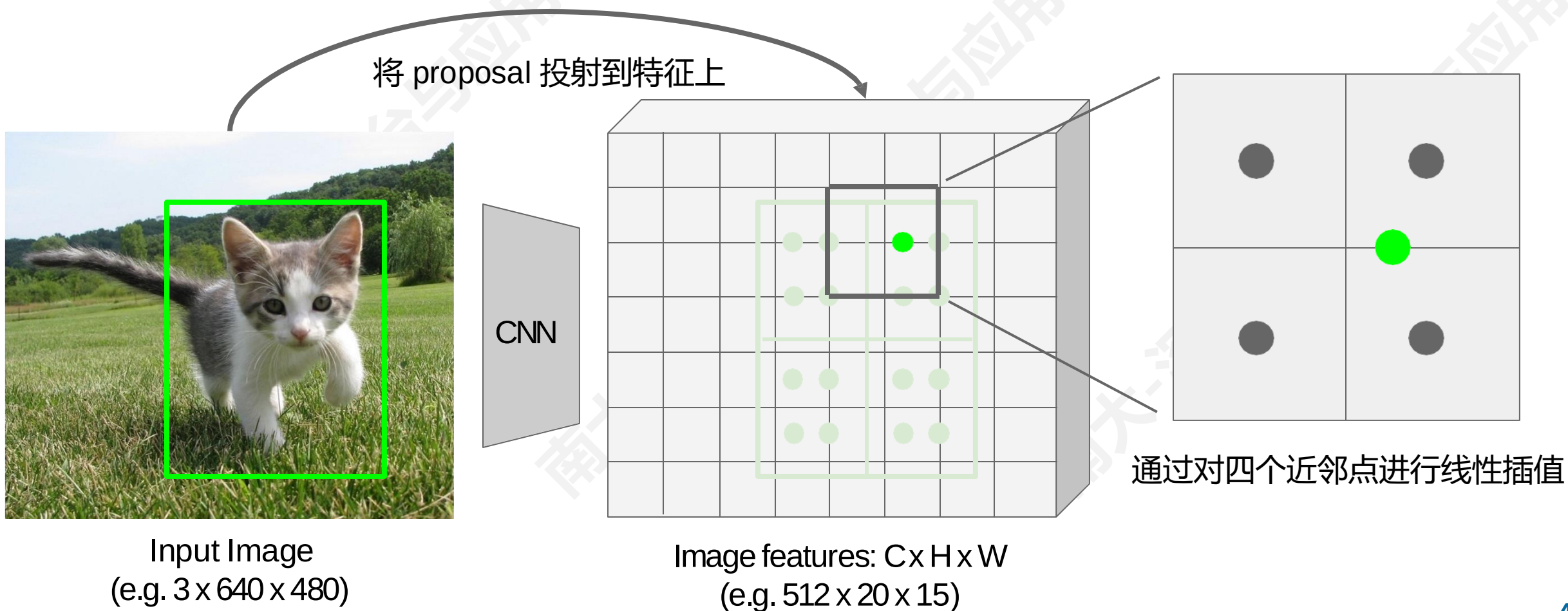
问题：RoI 提取的特征与原始 proposal 中的特征存在差异



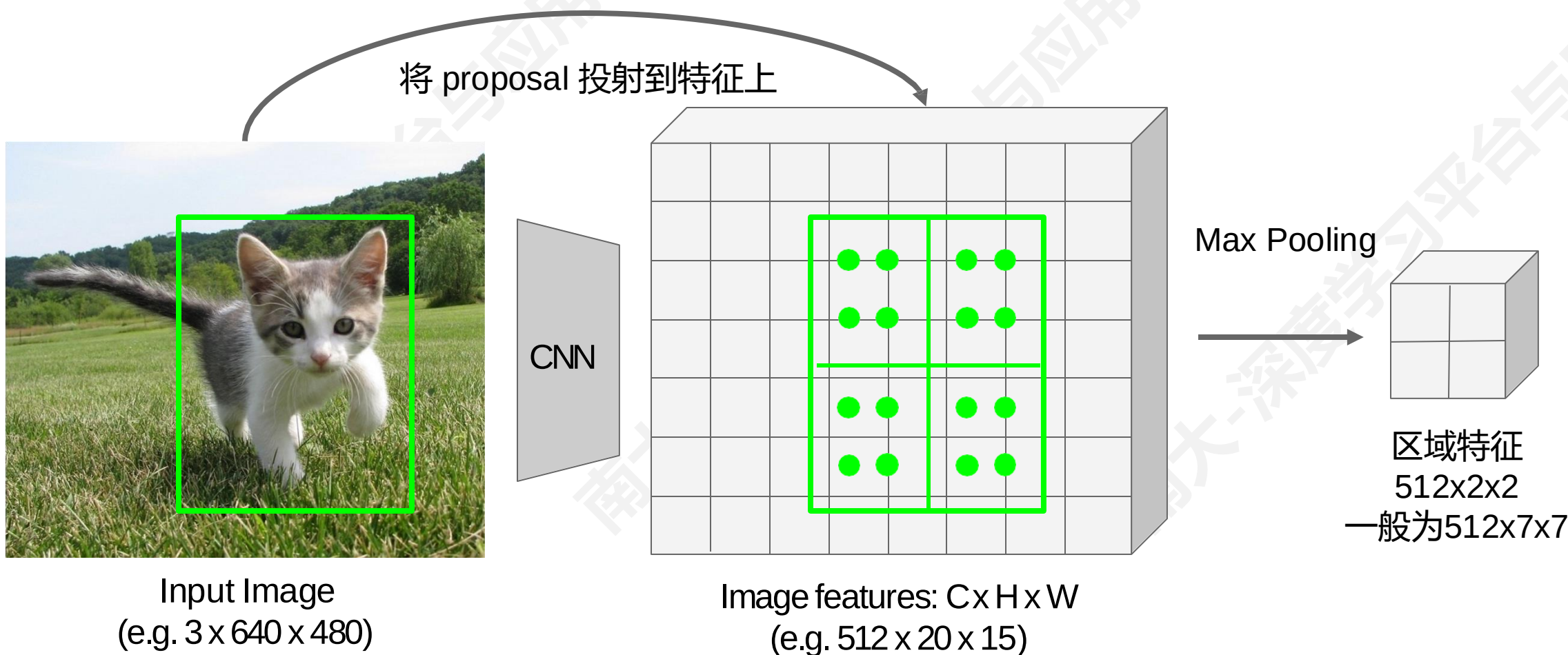
■ RoI Align



■ 计算机视觉任务

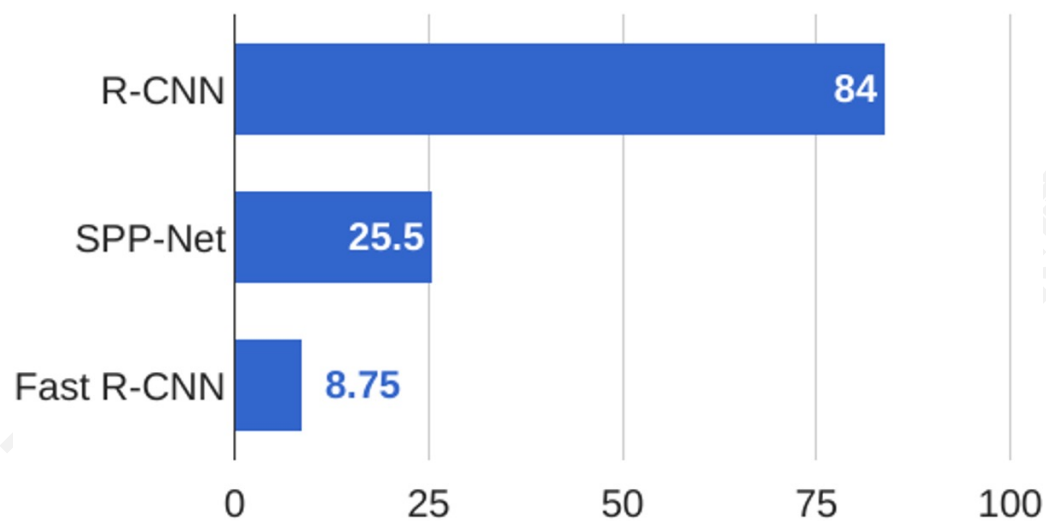


■ 计算机视觉任务

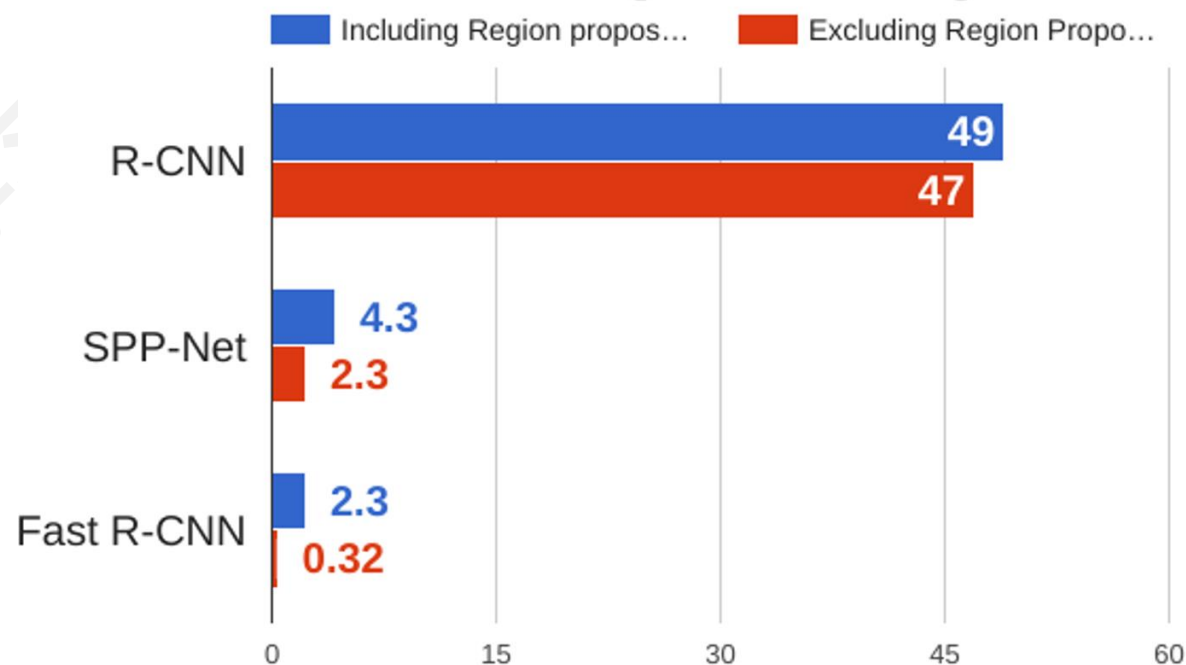


■ R-CNN v.s. Fast R-CNN

Training time (Hours)

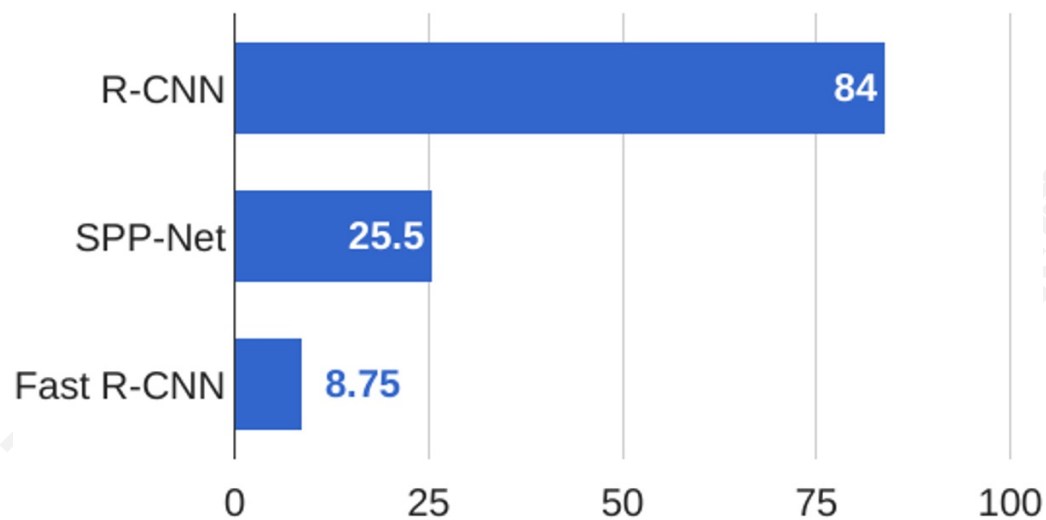


Test time (seconds)

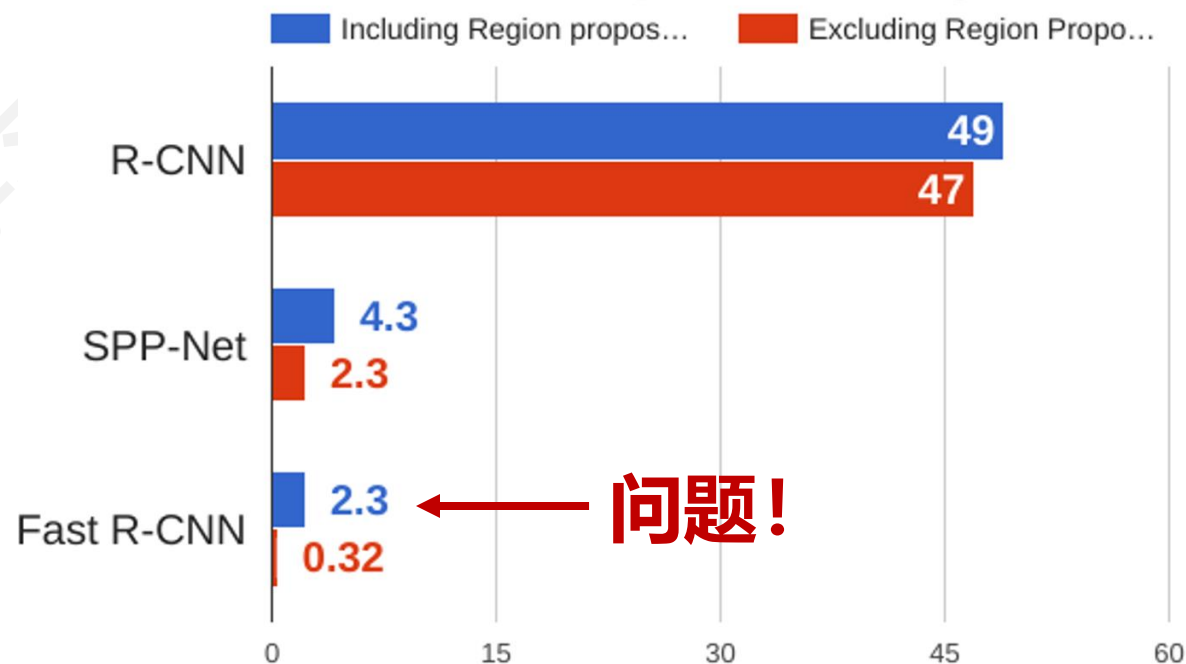


■ R-CNN v.s. Fast R-CNN

Training time (Hours)



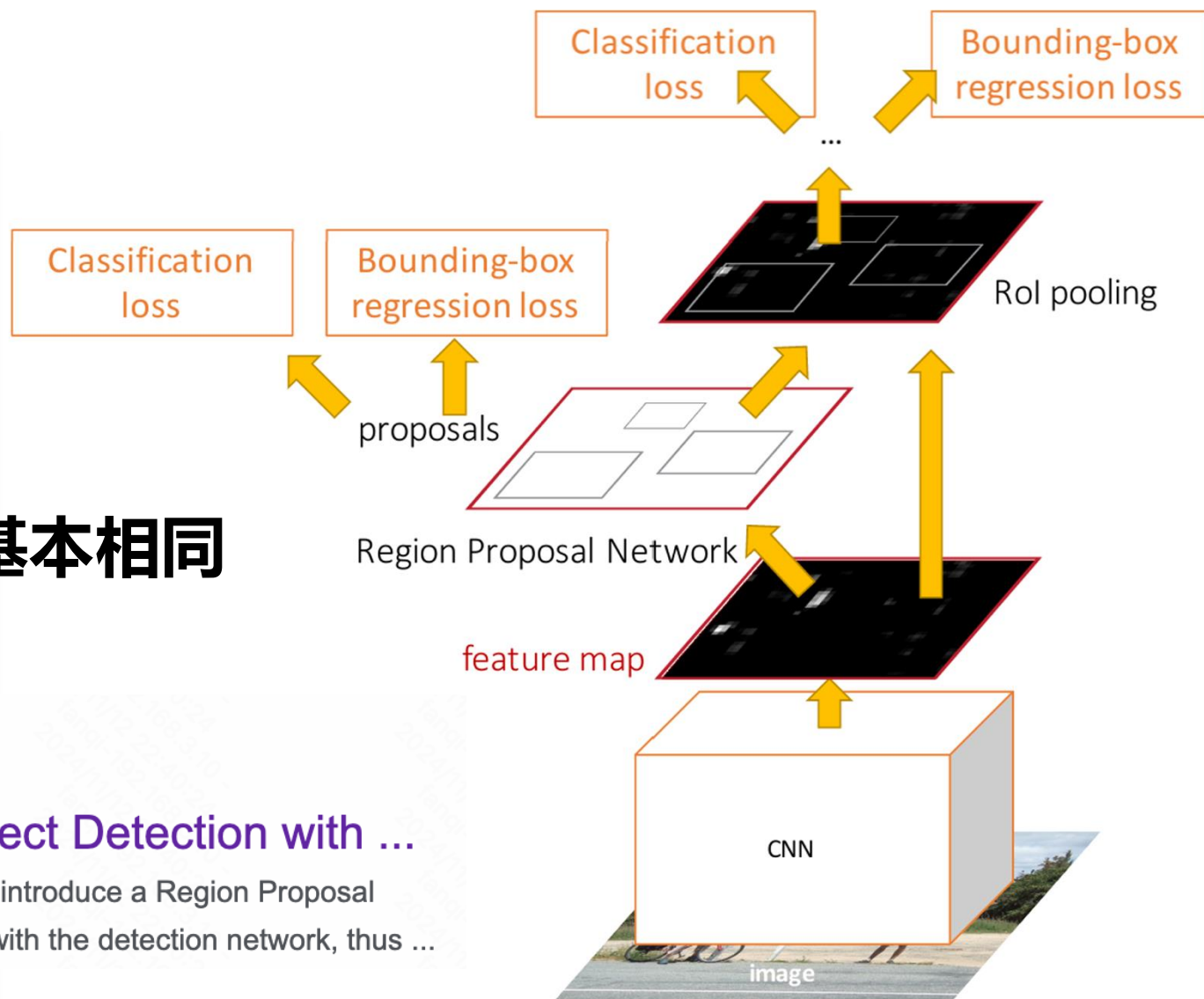
Test time (seconds)



问题!

■ Faster R-CNN

使用 CNN 预测 proposal!
其他模块与 Fast R-CNN 基本相同



arXiv

<https://arxiv.org> > cs · [翻译此页](#) ⋮

Faster R-CNN: Towards Real-Time Object Detection with ...

作者: S Ren · 2015 · 被引用次数: 50481 — In this work, we introduce a Region Proposal Network (RPN) that shares full-image convolutional features with the detection network, thus ...

■ Region Proposal Network



Input Image
(e.g. 3 x 640 x 480)

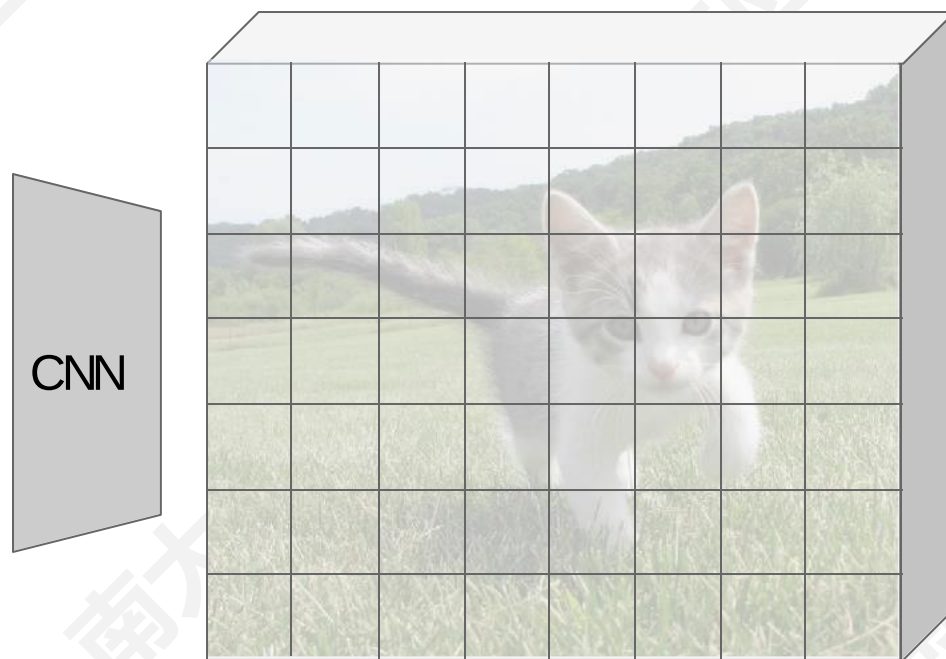


Image features
(e.g. 512 x 20 x 15)

■ Region Proposal Network

假设每个点都有固定大小的锚框



Input Image
(e.g. 3 x 640 x 480)

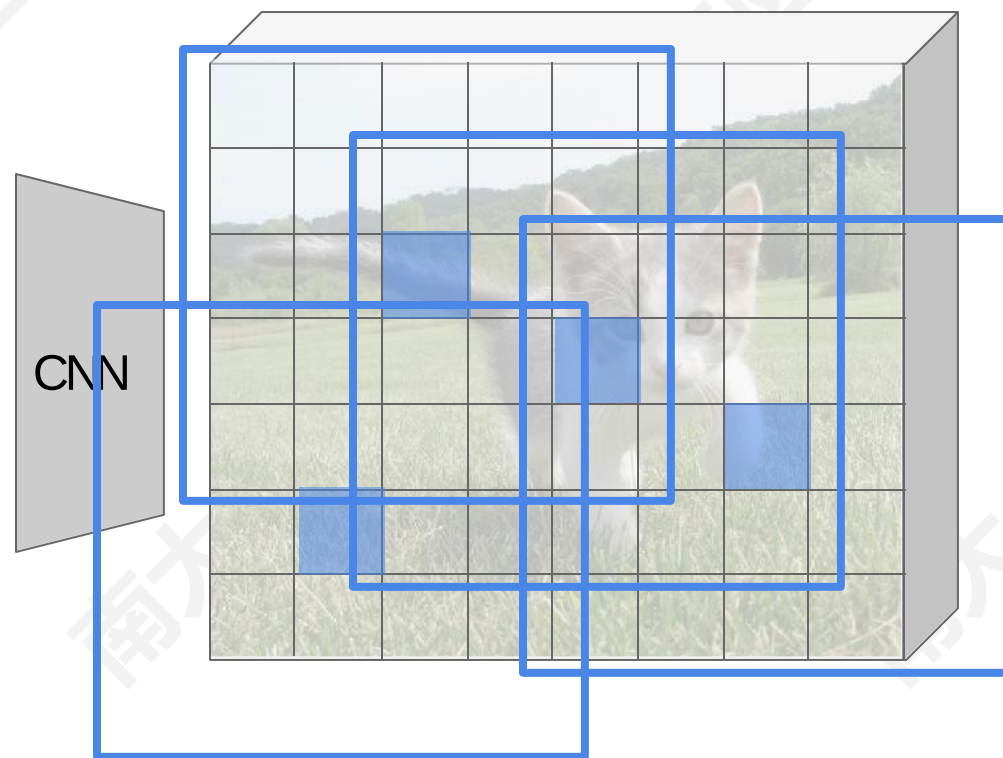


Image features
(e.g. 512 x 20 x 15)

■ Region Proposal Network

假设每个点都有固定大小的锚框



Input Image
(e.g. 3 x 640 x 480)

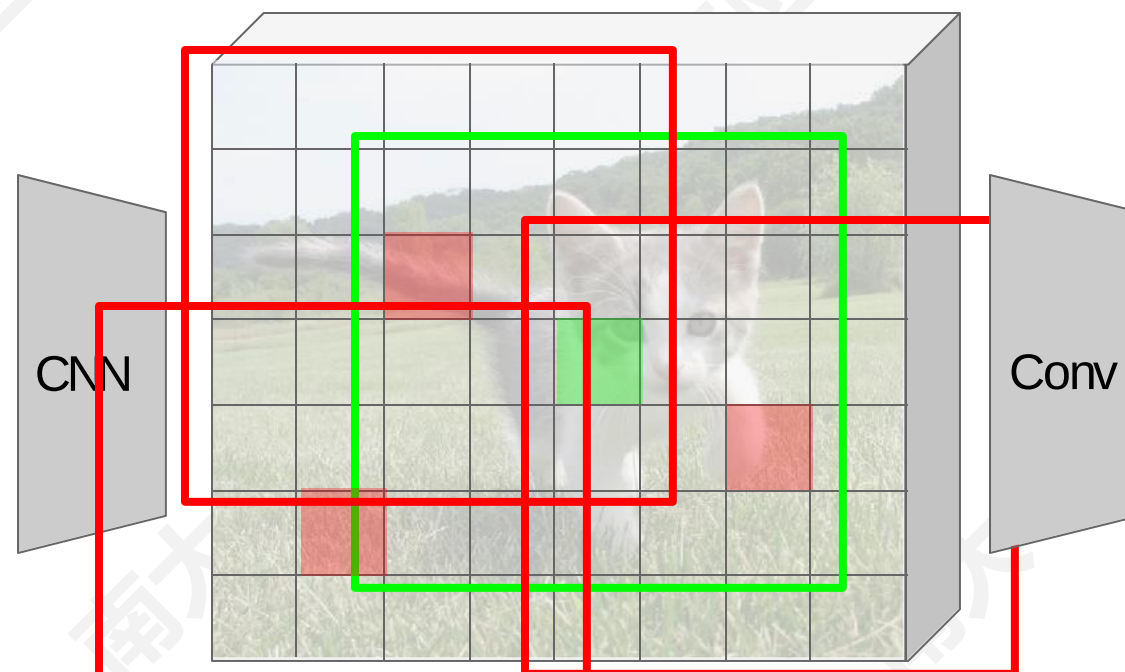


Image features
(e.g. 512 x 20 x 15)

Anchor is an object?
1 x 20 x 15

预测每个锚框内是否存在物体

■ Region Proposal Network

预测每个锚框内是否存在物体

对于存在物体的锚框，
对锚框进行调整



Input Image
(e.g. 3 x 640 x 480)

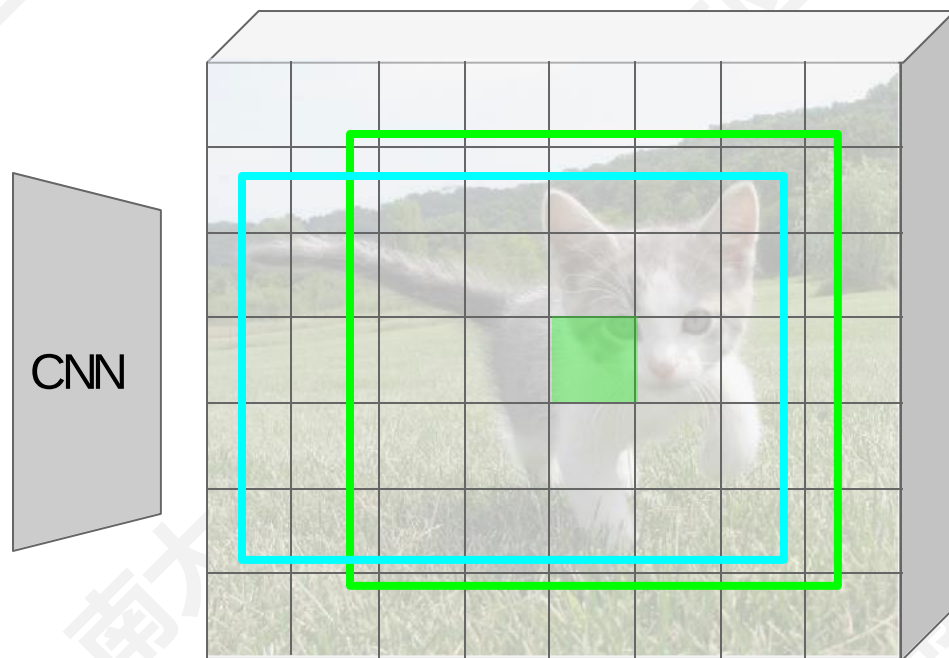


Image features
(e.g. 512 x 20 x 15)



Anchor is an object?
1 x 20 x 15

Box corrections
4 x 20 x 15

■ Region Proposal Network

一般来说，每个点会定义 K 个锚框，具有不同的大小和比例



Input Image
(e.g. $3 \times 640 \times 480$)

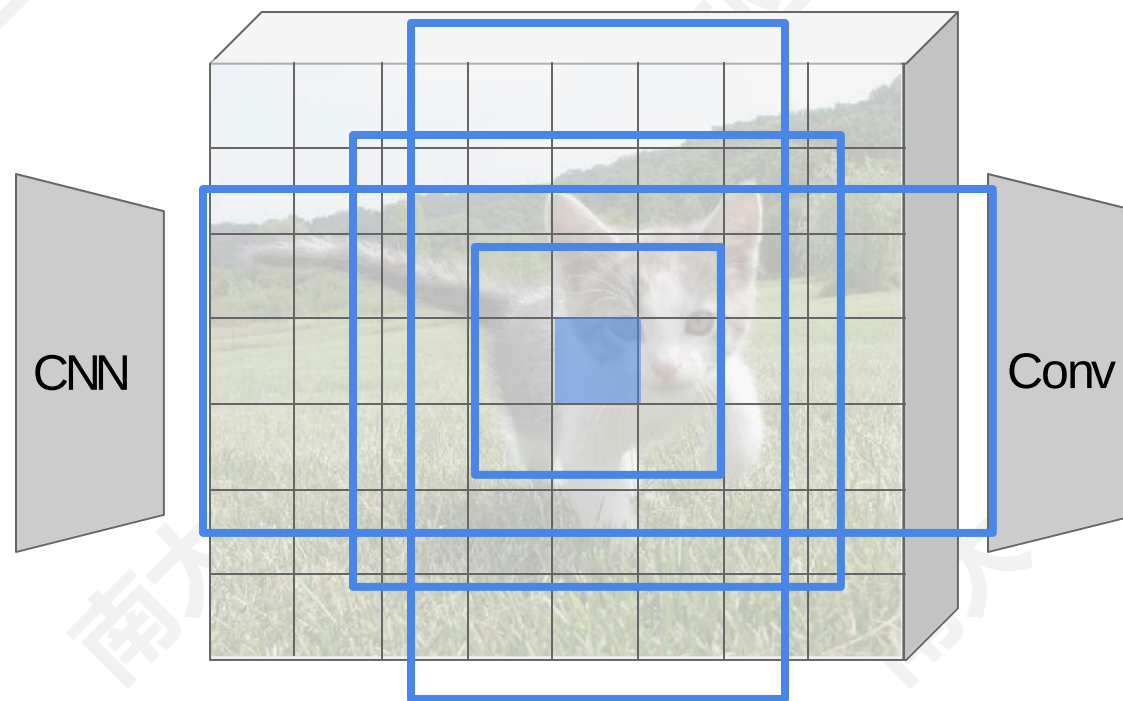


Image features
(e.g. $512 \times 20 \times 15$)

Anchor is an object?
 $K \times 20 \times 15$

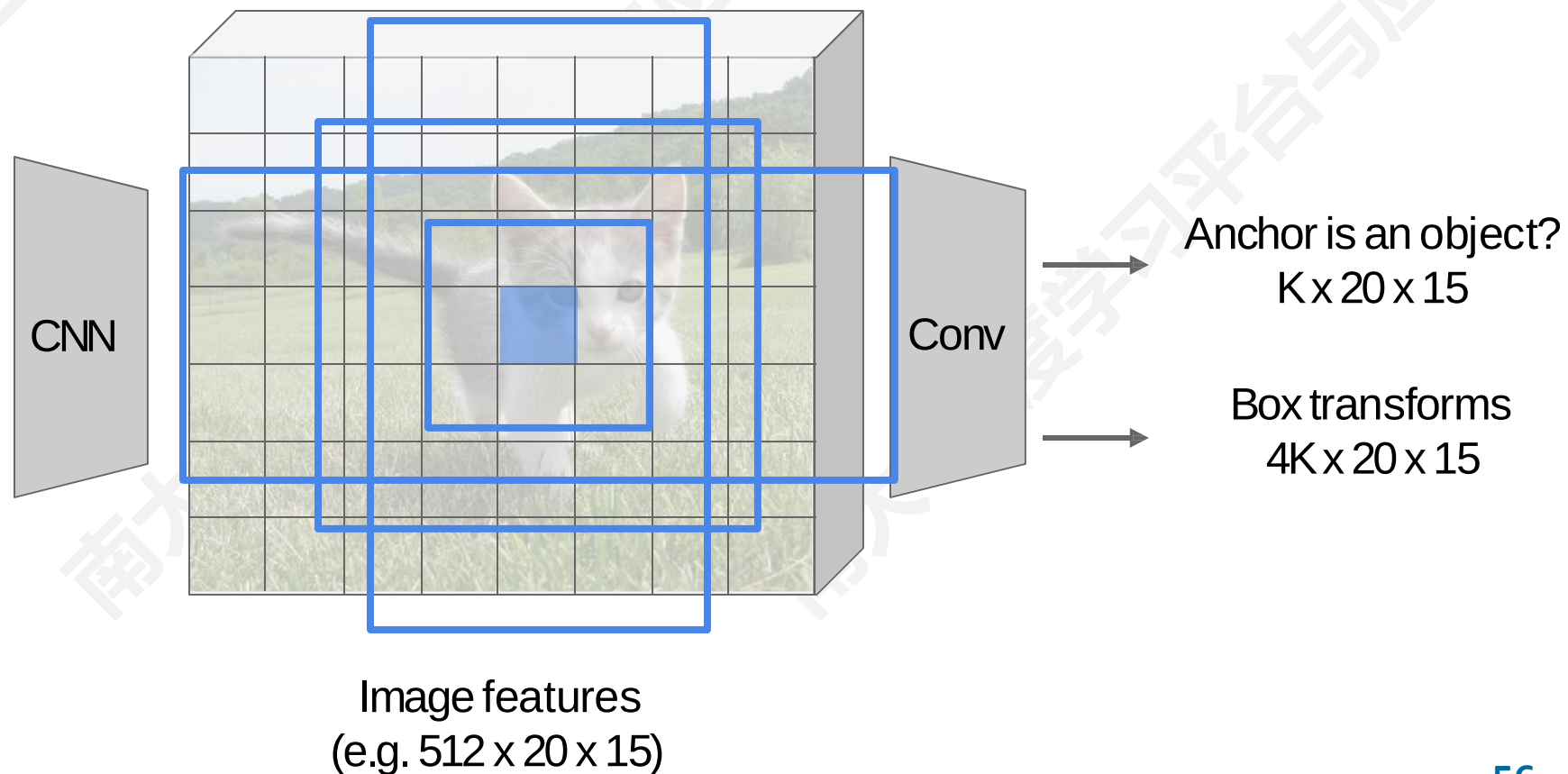
Box transforms
 $4K \times 20 \times 15$

■ Region Proposal Network

对 $K \times 20 \times 15$ 个锚框预测存在物体的可能性，选择 top300 作为 proposal



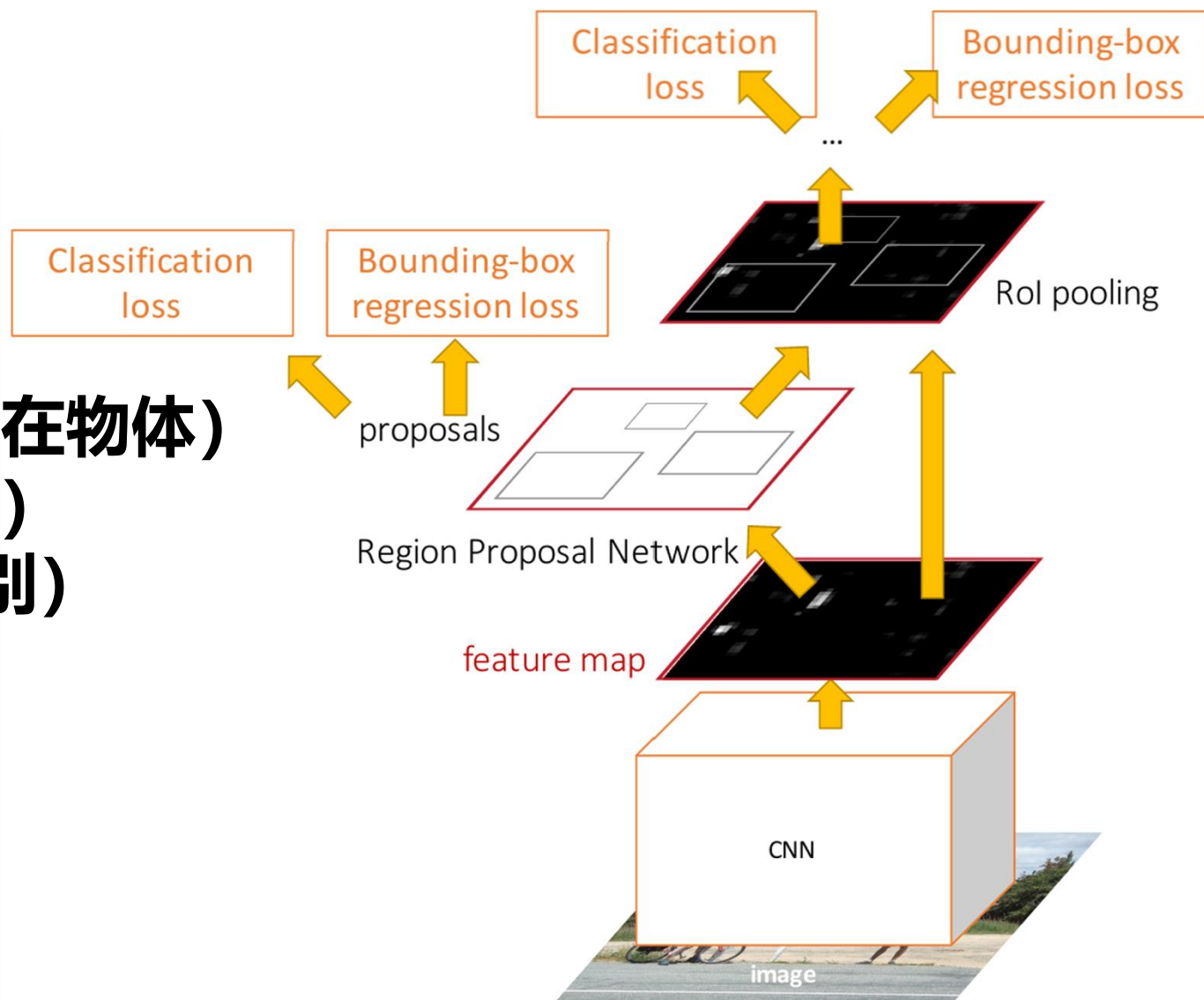
Input Image
(e.g. $3 \times 640 \times 480$)



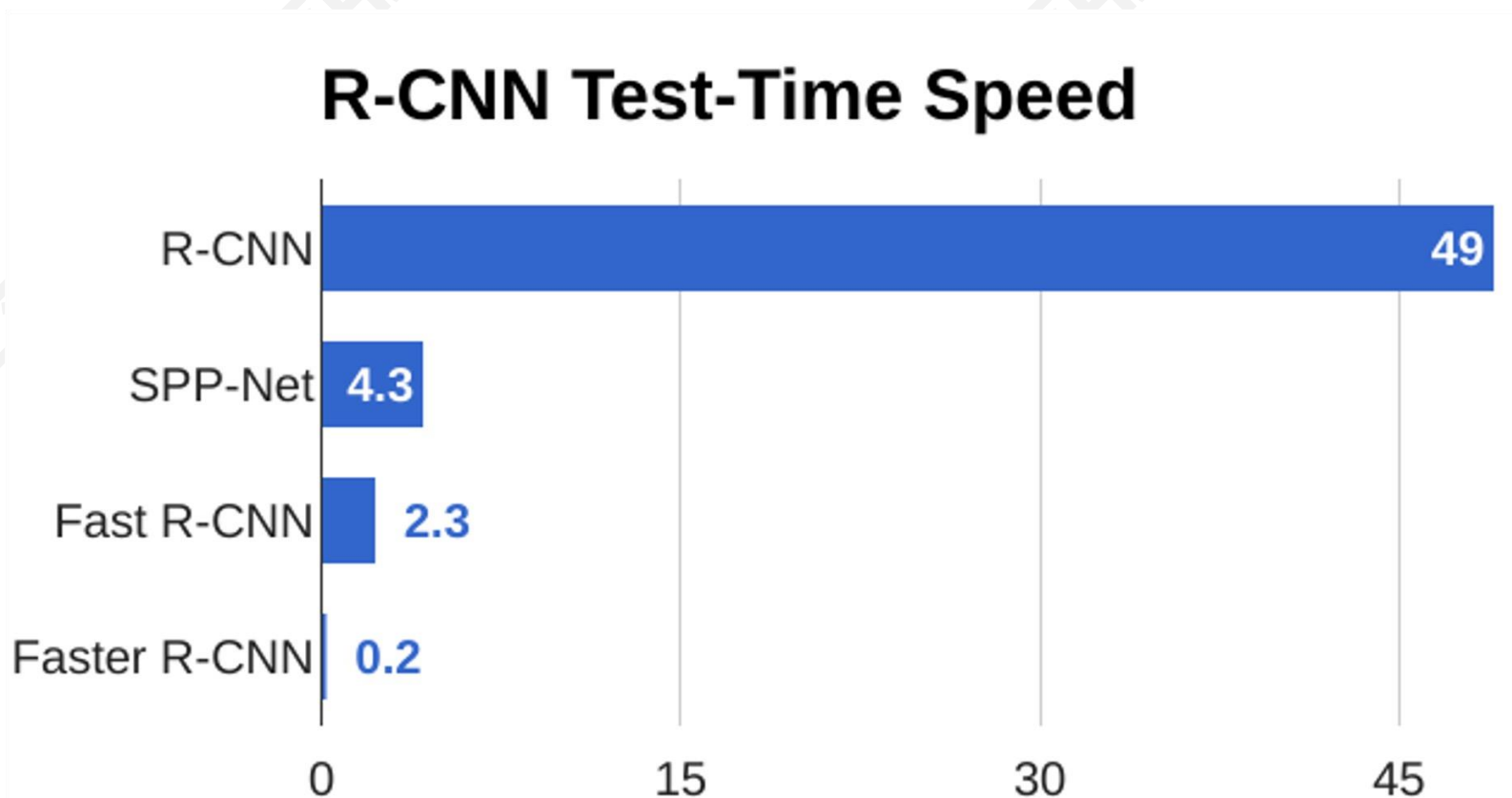
■ Faster R-CNN

联合训练4个损失:

- RPN 分类损失 (是否存在物体)
- RPN 回归损失 (矩形框)
- 检测分类损失 (物体类别)
- 检测回归损失 (矩形框)



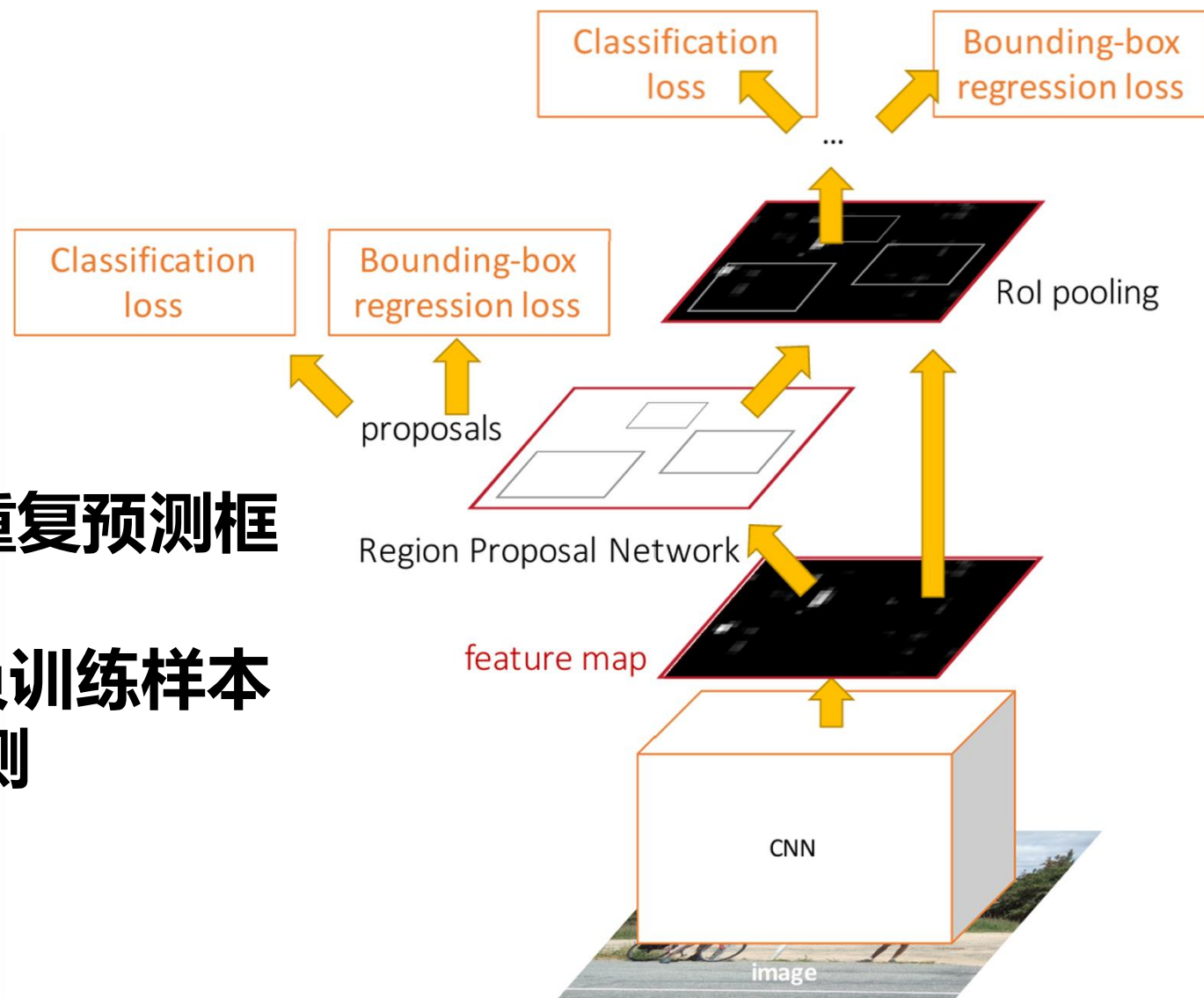
■ Faster R-CNN



■ Faster R-CNN

更多细节:

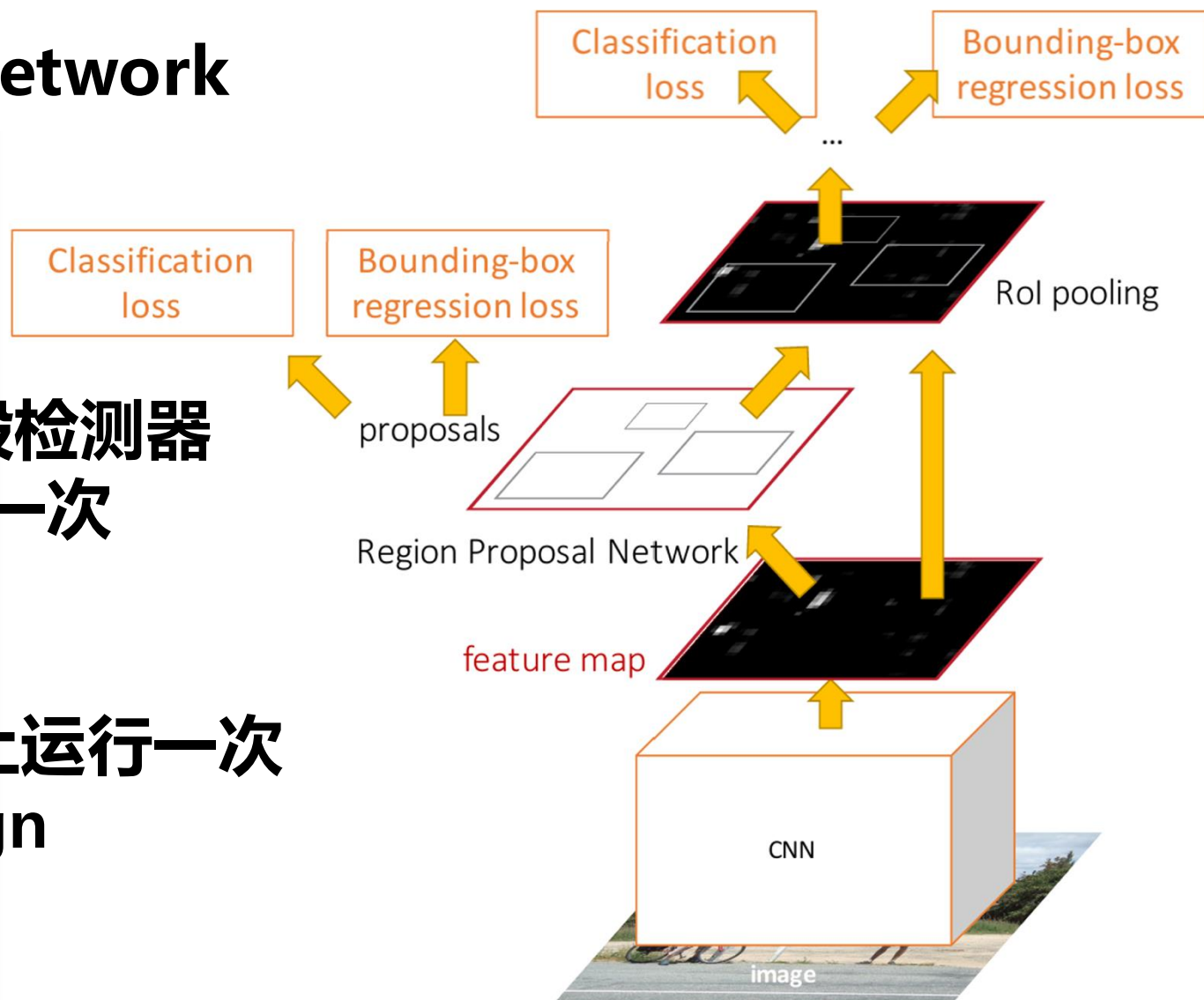
- 如何使用 NMS 去除重复预测框
- 如何确定锚框
- 如何为 RPN 采样正负训练样本
- 如何进行矩形框的预测



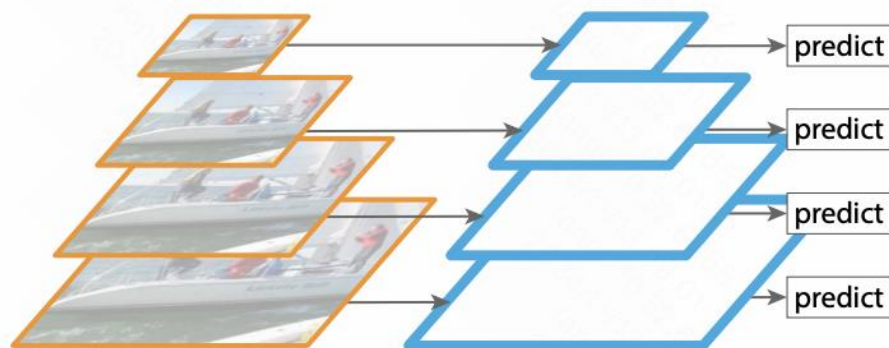
■ Region Proposal Network

Faster R-CNN 是一个两阶段检测器

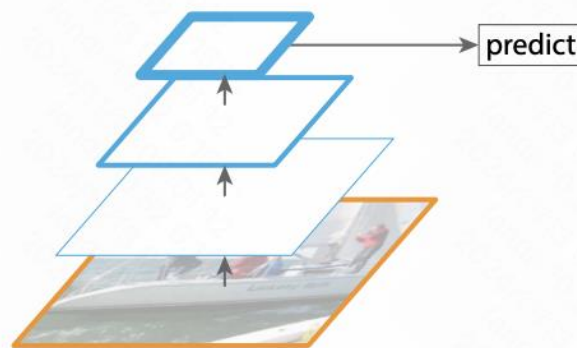
- 第一阶段：每张图上运行一次
 - Backbone
 - RPN
- 第二阶段：每个 region 上运行一次
 - RoI Pooling / RoI Align
 - Detector



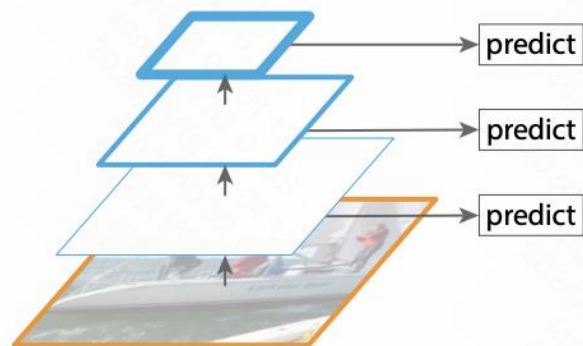
■ Feature Pyramid Network



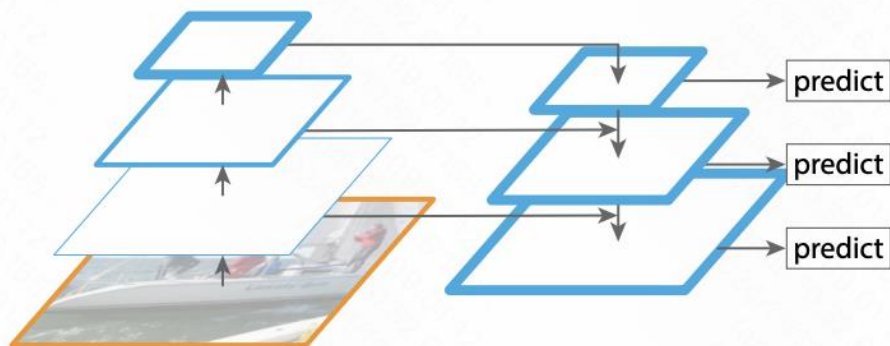
(a) Featurized image pyramid



(b) Single feature map



(c) Pyramidal feature hierarchy



(d) Feature Pyramid Network

■ Cascade R-CNN

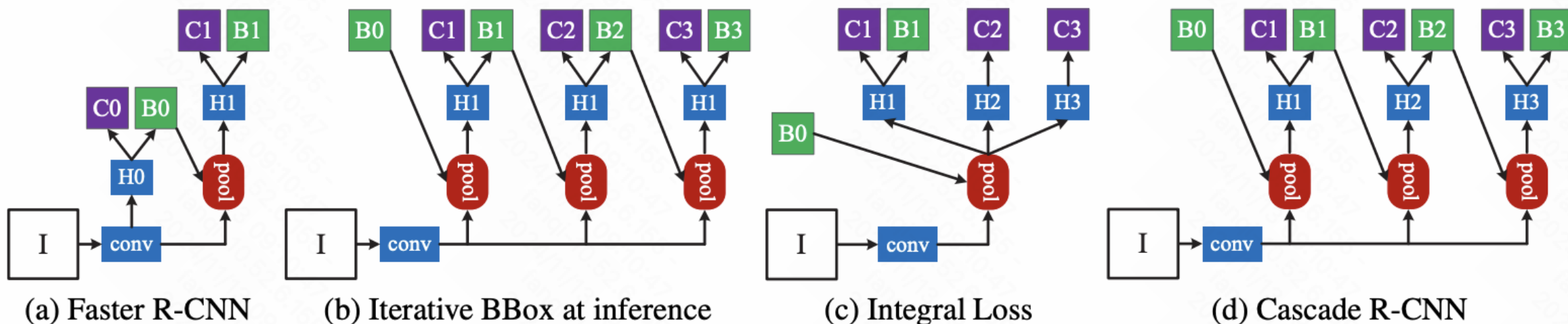
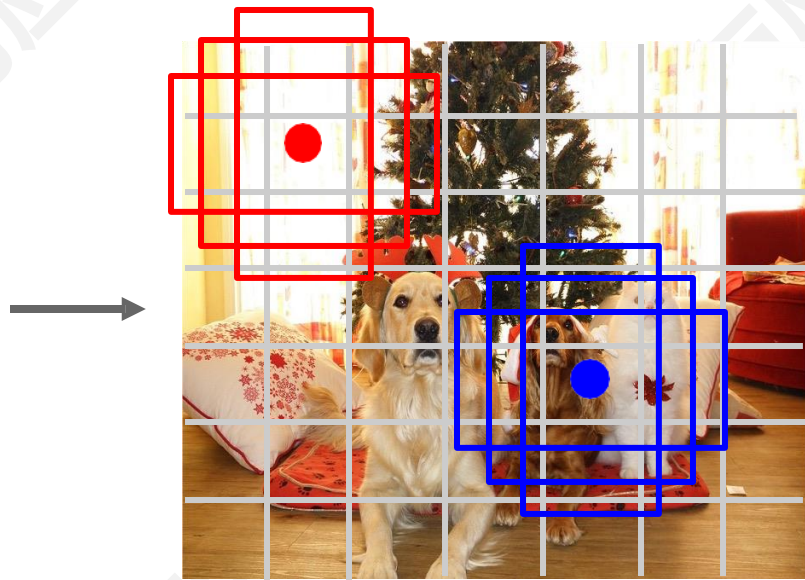


Figure 3. The architectures of different frameworks. “I” is input image, “conv” backbone convolutions, “pool” region-wise feature extraction, “H” network head, “B” bounding box, and “C” classification. “B0” is proposals in all architectures.

■ 单阶段检测器：YOLO/SSD/RetinaNet



Input image
 $3 \times H \times W$



Divide image into grid
 7×7

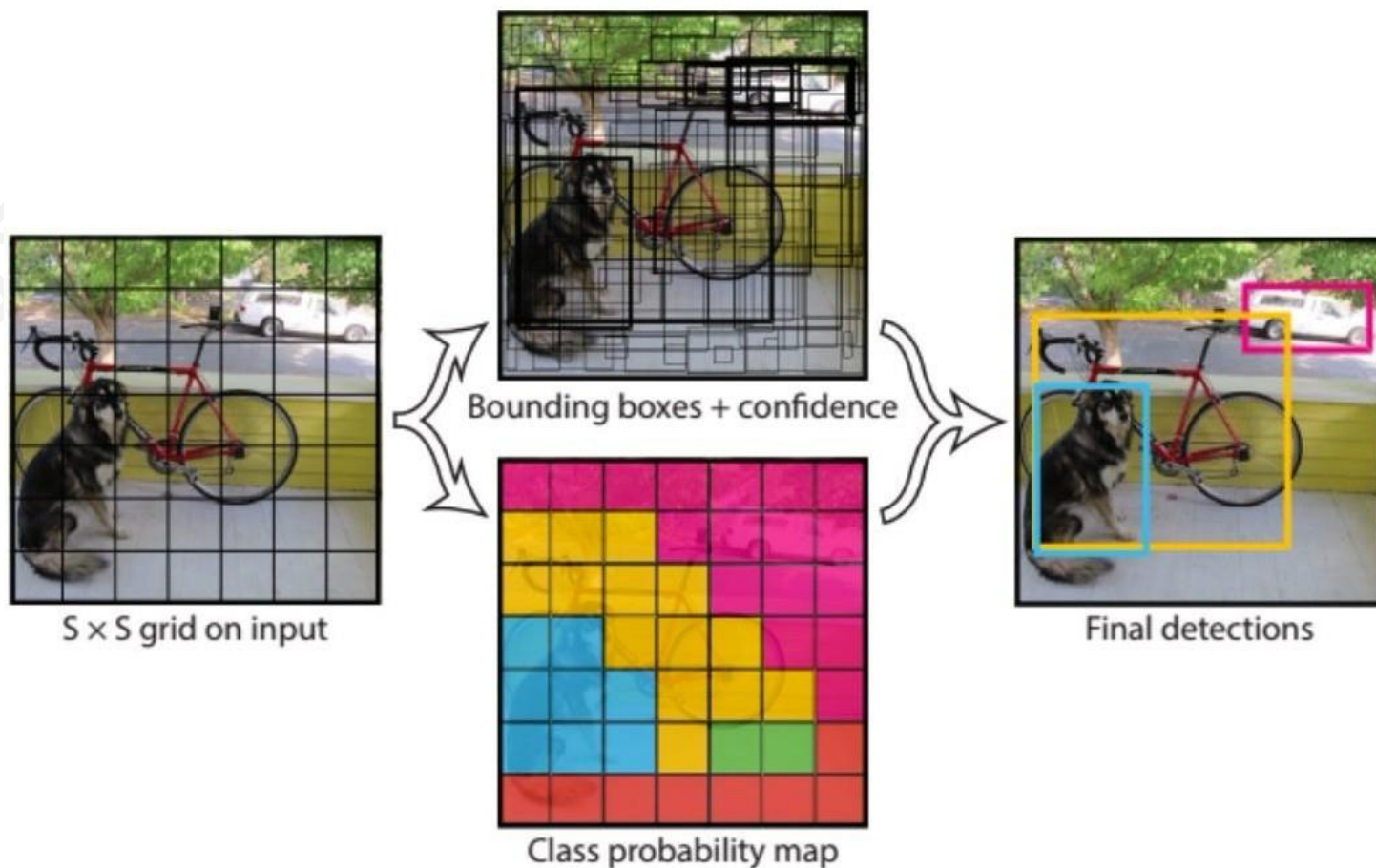
直接对每个锚框
进行分类和回归

可以看作是一个
有类别的 RPN

You Only Look Once: Unified, Real-Time Object Detection

作者: J Redmon · 2015 · 被引用次数: 54035 — We present **YOLO**, a new approach to object detection. Prior work on object detection repurposes classifiers to perform detection.

■ YOLO-You Only Look Once



■ Fully Convolutional One-Stage Object Detection

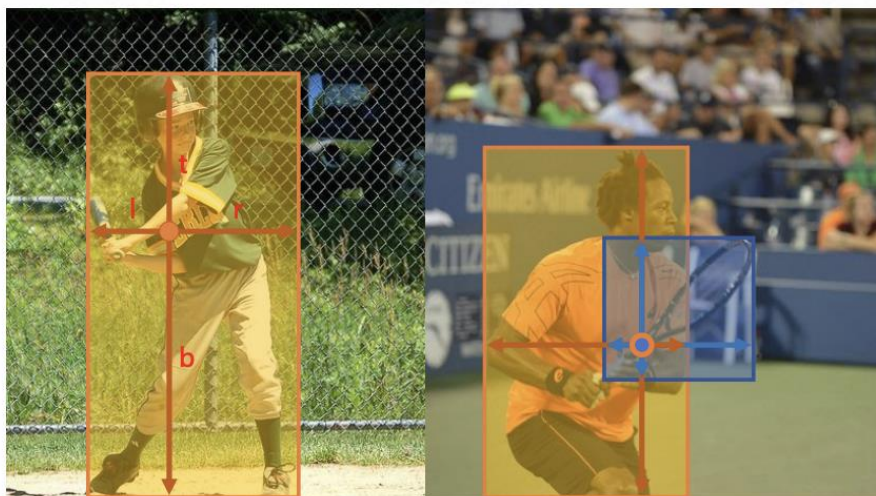
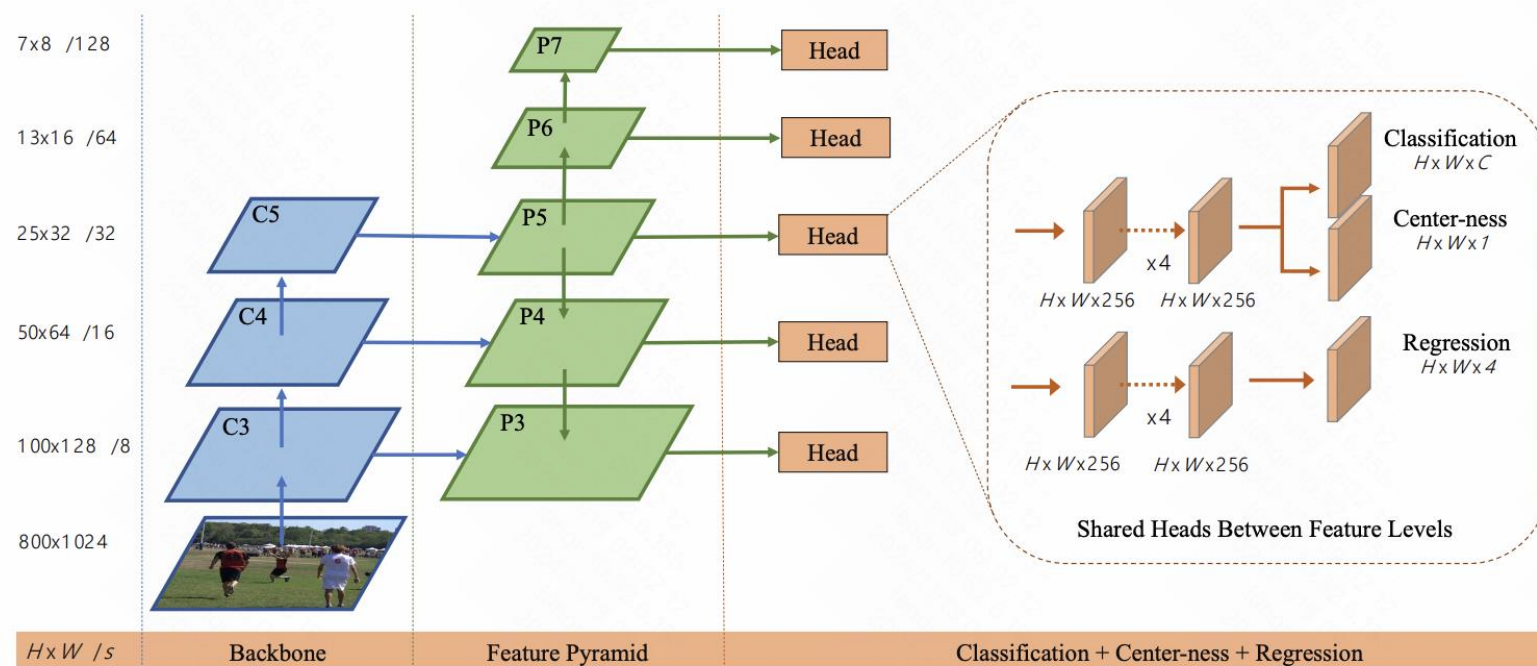
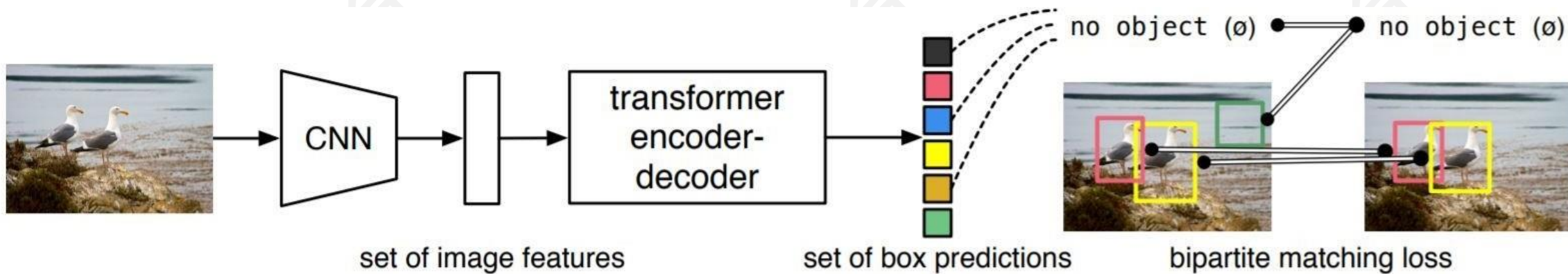


Figure 1 – As shown in the left image, FCOS works by predicting a 4D vector (l, t, r, b) encoding the location of a bounding box at each foreground pixel (supervised by ground-truth bounding box information during training). The right plot shows that when a location residing in multiple bounding boxes, it can be ambiguous in terms of which bounding box this location should regress.



■ DETR (DEtection TRansformer)



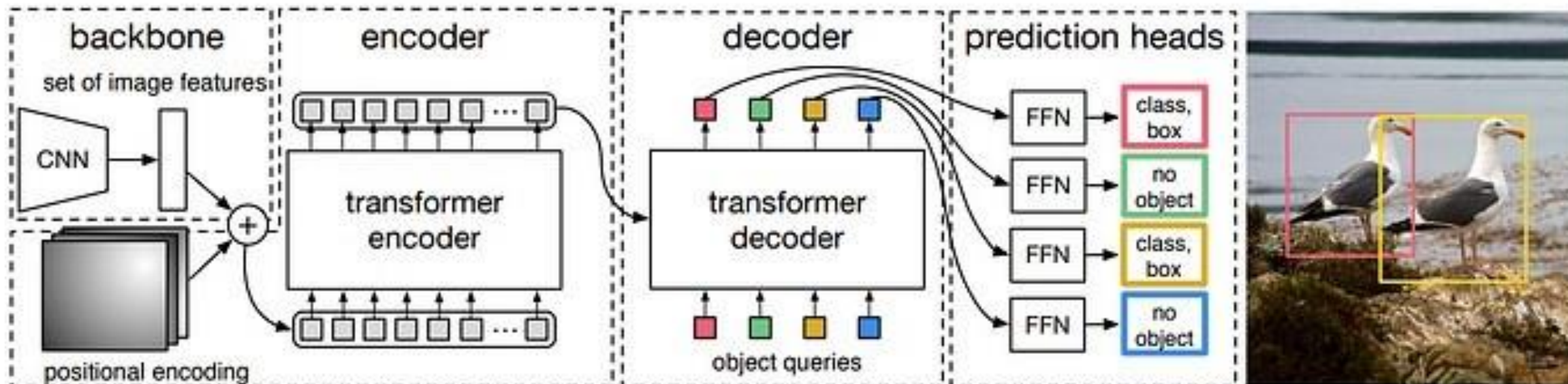
arXiv

<https://arxiv.org> > cs · [翻译此页](#) :

[2005.12872] End-to-End Object Detection with Transformers

作者: N Carion · 2020 · 被引用次数: 14361 — DETR demonstrates accuracy and run-time performance on par with the well-established and highly-optimized Faster RCNN baseline on...

■ DETR (DEtection TRansformer)



大 纲

- 语义分割
- 目标检测
- 实例分割

■ 实例分割-对检测到的每个实例物体进行分割

Classification



CAT

No spatial extent

Semantic
Segmentation



GRASS, CAT, TREE, SKY

No objects, just pixels

Object
Detection



DOG, DOG, CAT

Instance
Segmentation



DOG, DOG, CAT

Multiple Object

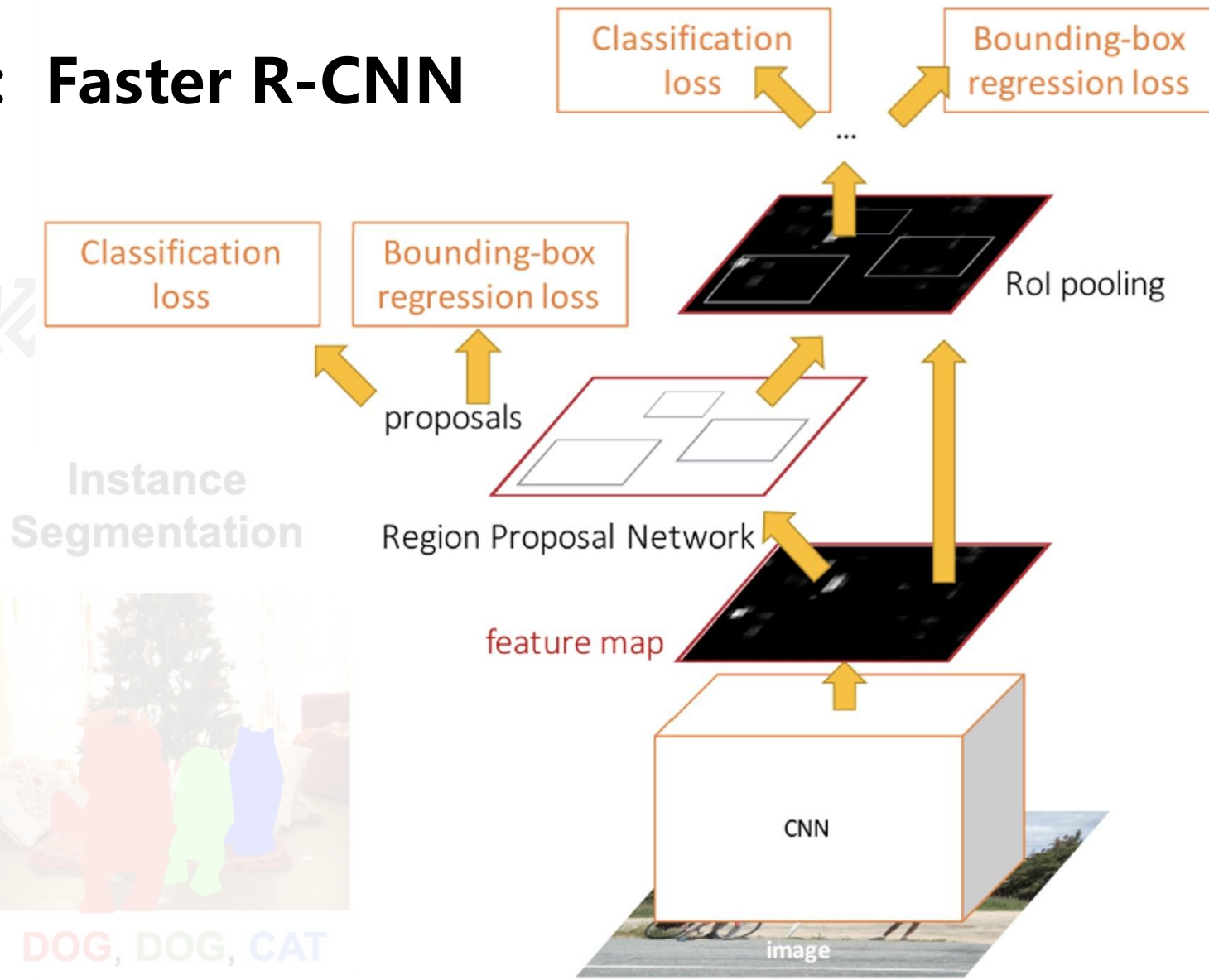
■ 目标检测：Faster R-CNN



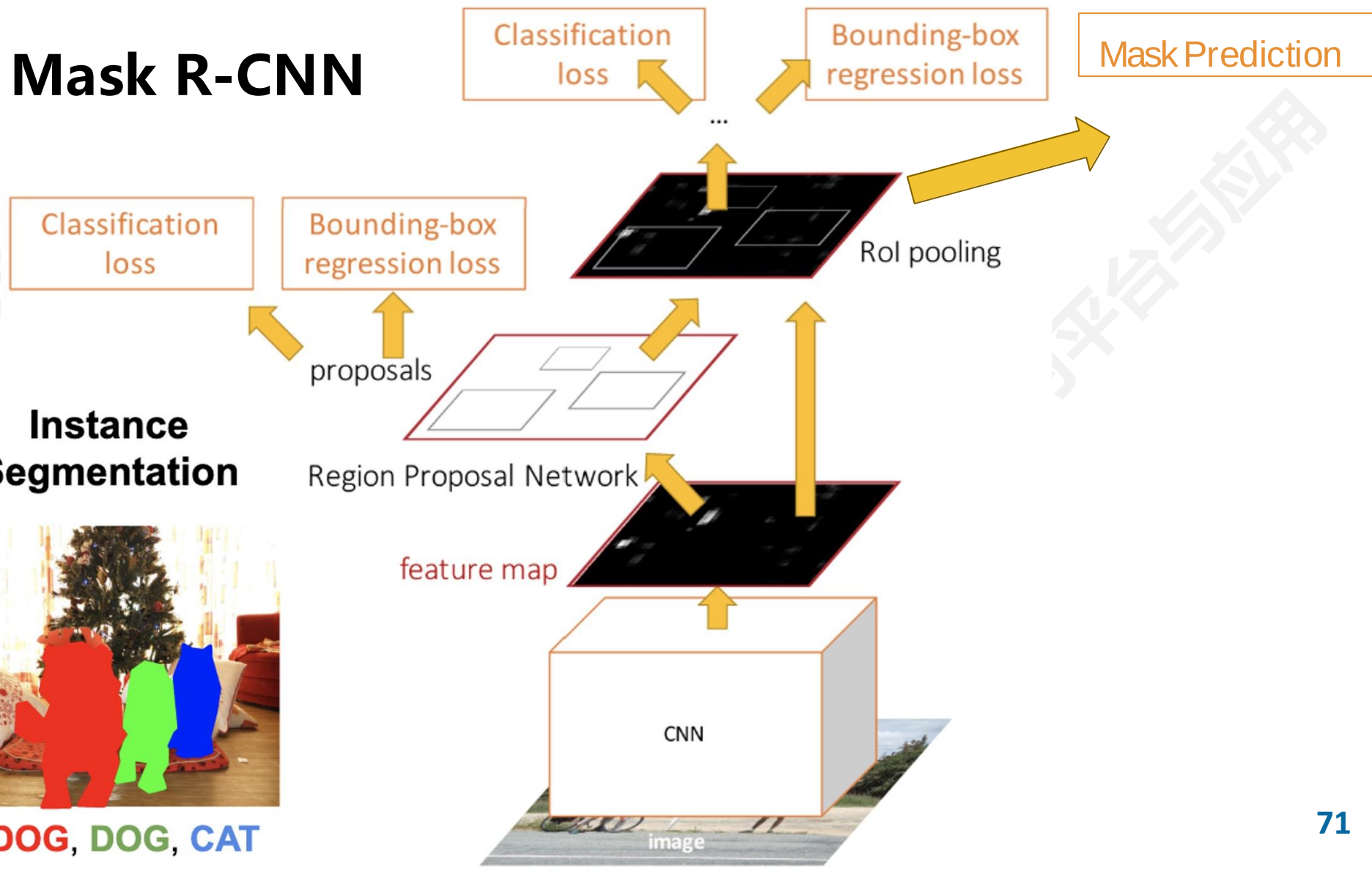
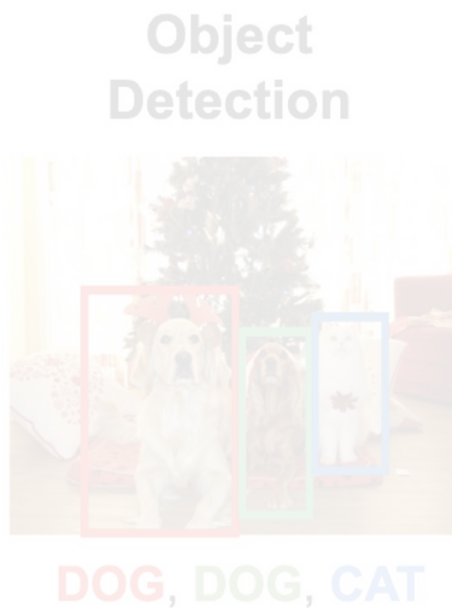
DOG, DOG, CAT



DOG, DOG, CAT



■ 实例分割：Mask R-CNN



实例分割



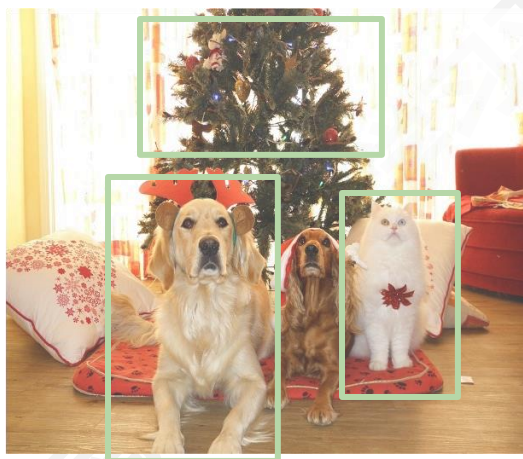
arXiv

<https://arxiv.org> > cs · [翻译此页](#) :

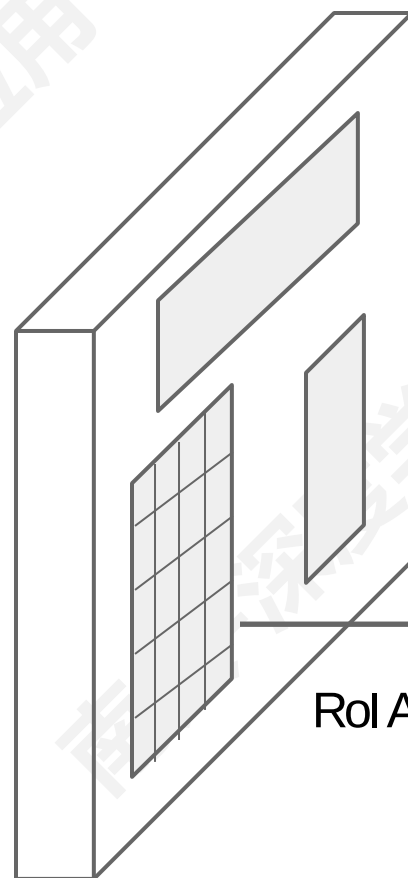
[1703.06870] Mask R-CNN

作者: K He · 2017 · 被引用次数: 39045 — We present a conceptually simple, flexible, and general framework for object instance segmentation. Our approach efficiently detects objects i...

■ Mask R-CNN



CNN
+RPN



RoI Align

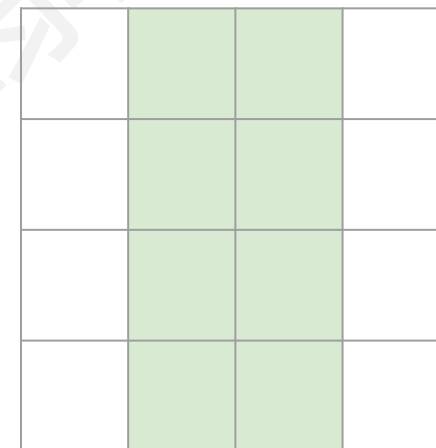
$256 \times 14 \times 14$

Classification Scores: C
Box coordinates (per class): $4 * C$

Conv

$256 \times 14 \times 14$

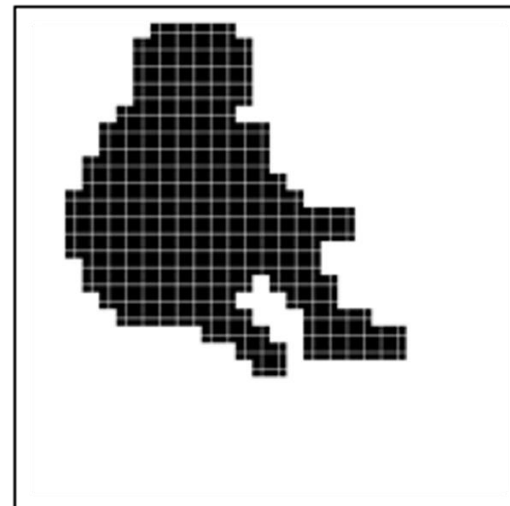
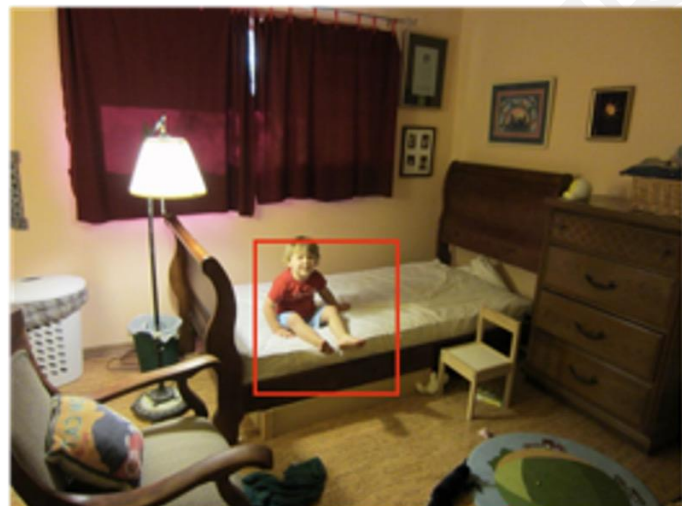
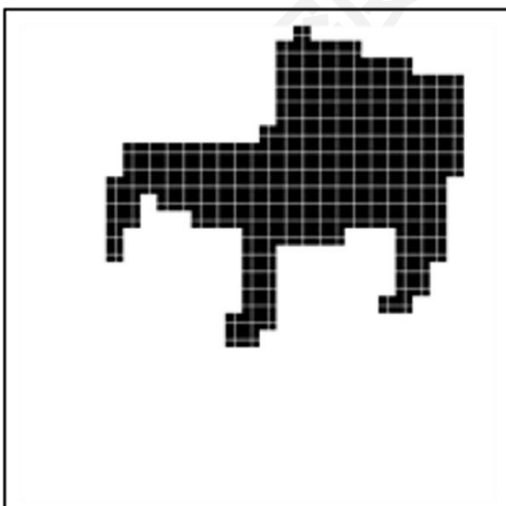
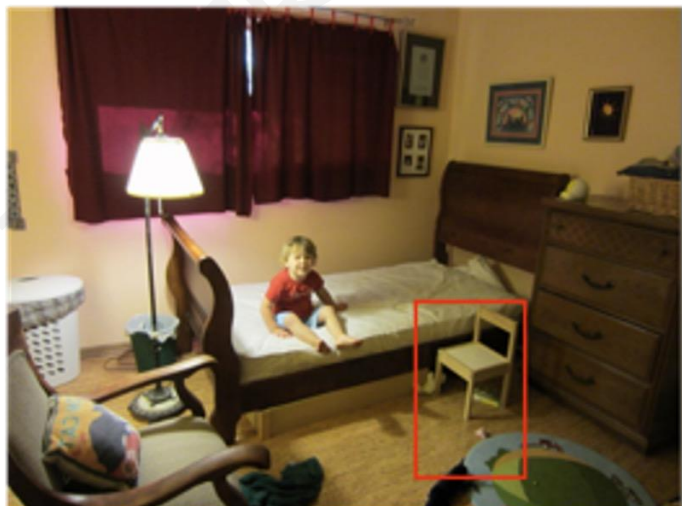
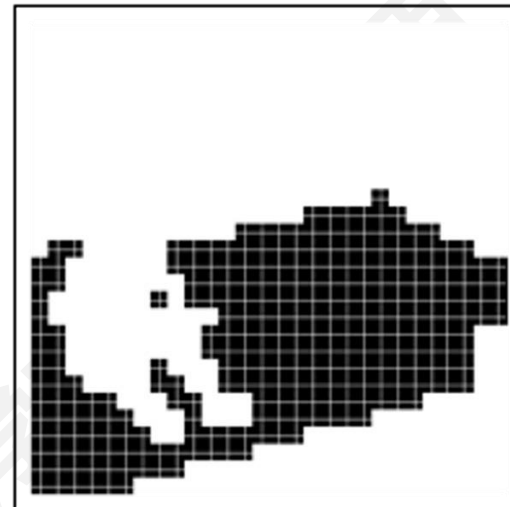
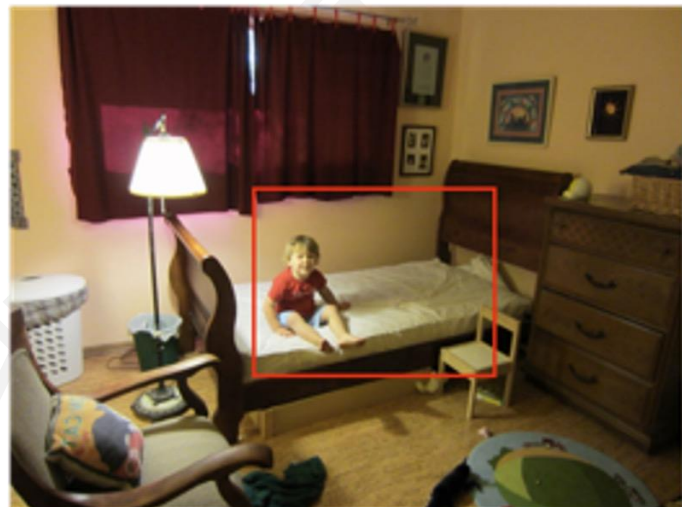
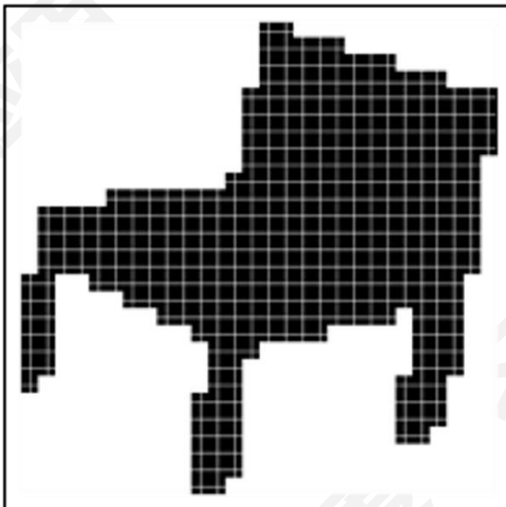
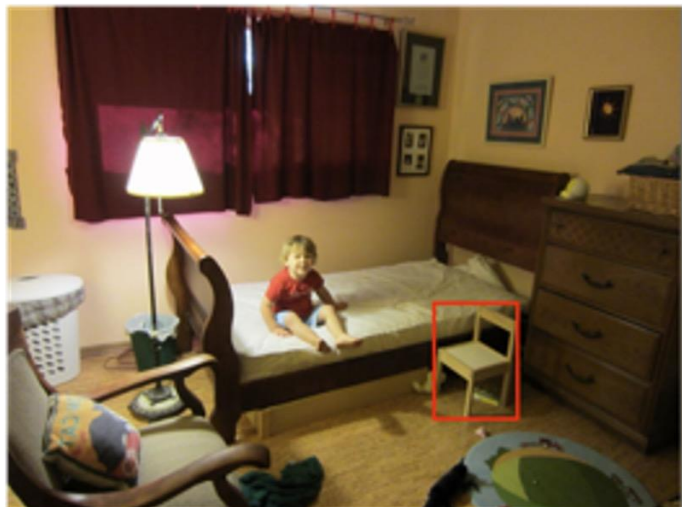
Conv



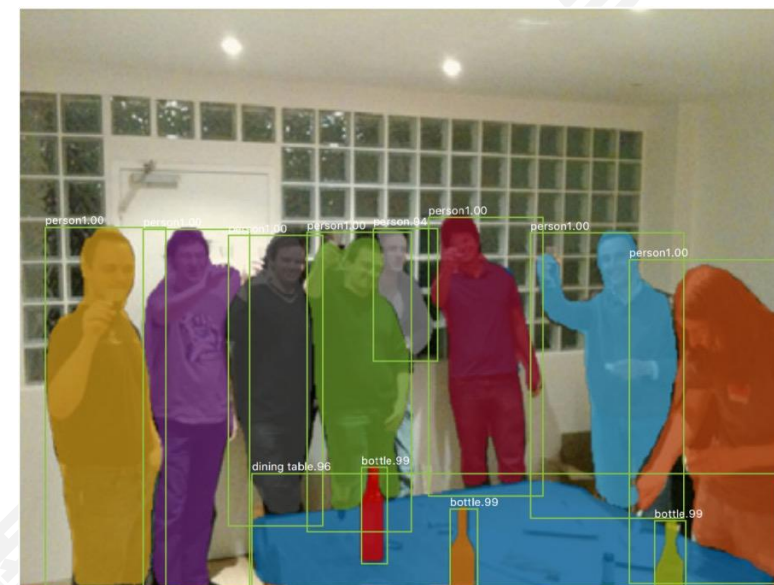
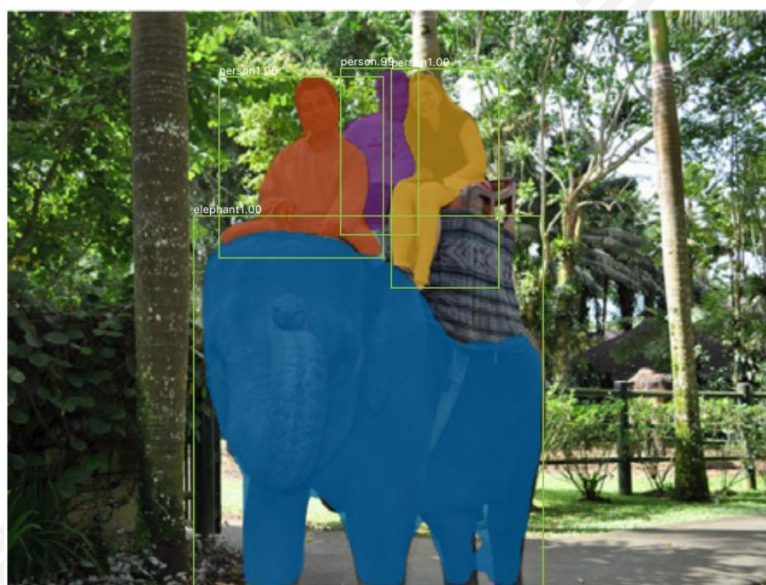
Predict a mask for
each of C classes

$C \times 28 \times 28$

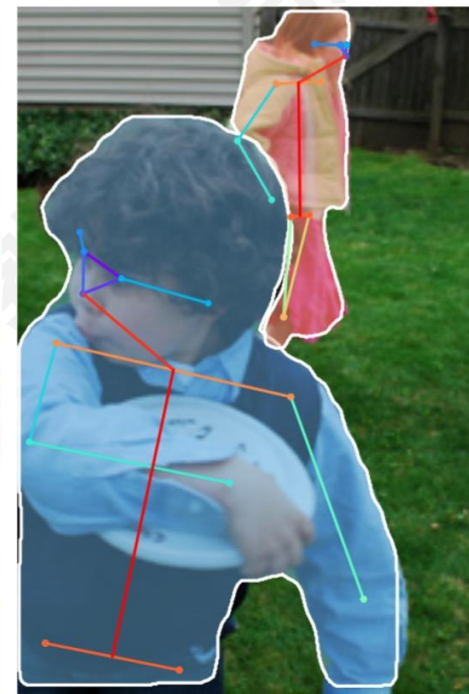
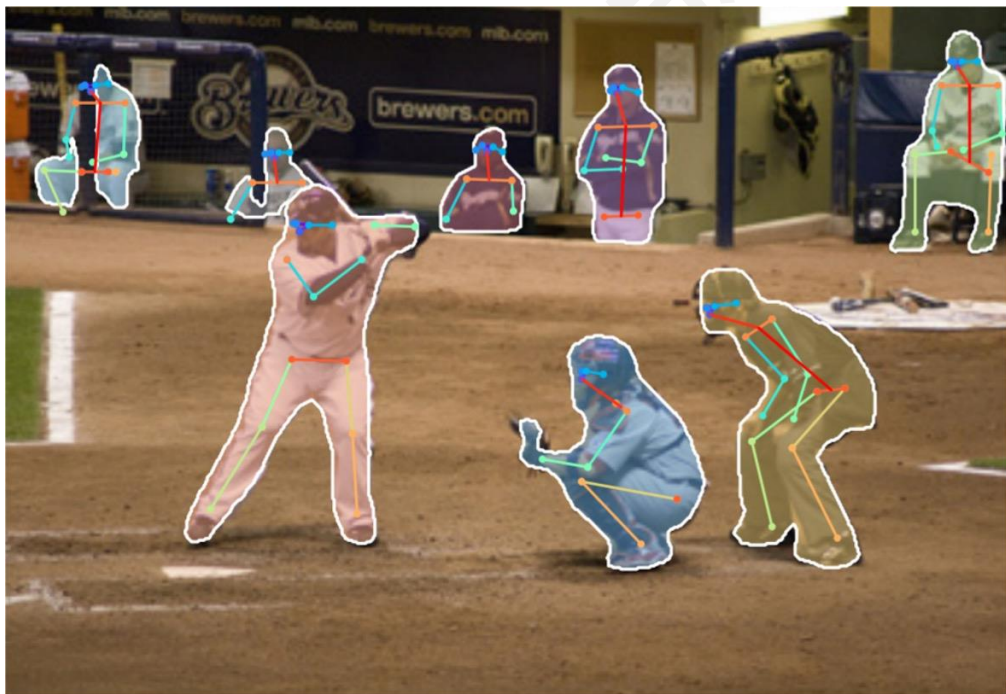
■ Mask R-CNN



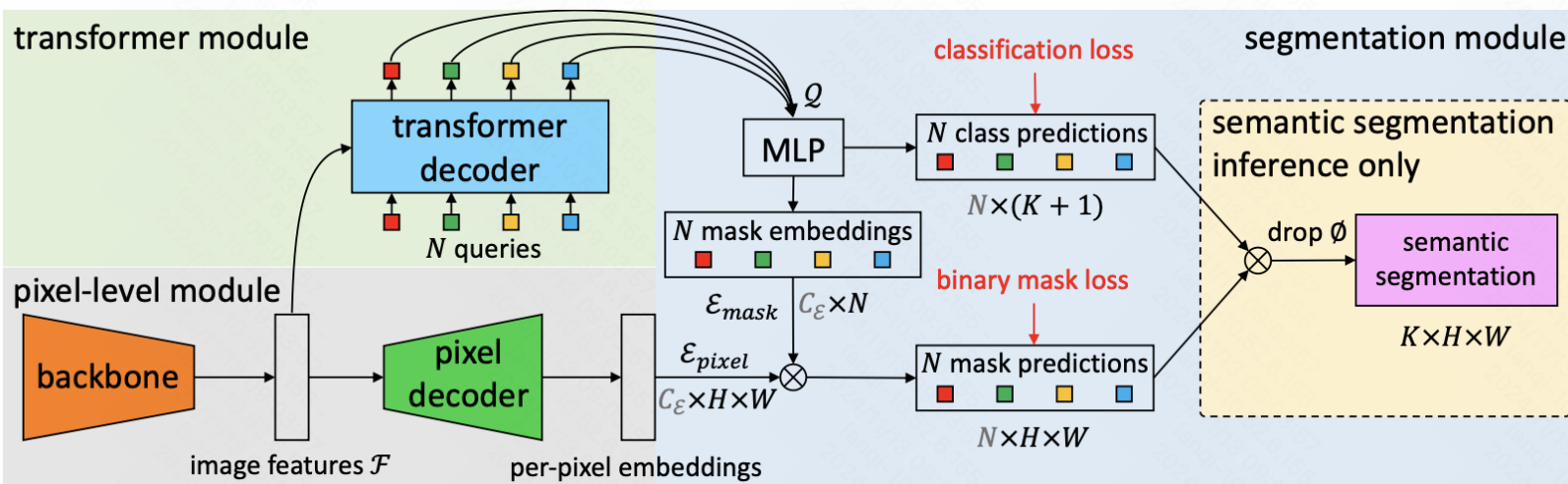
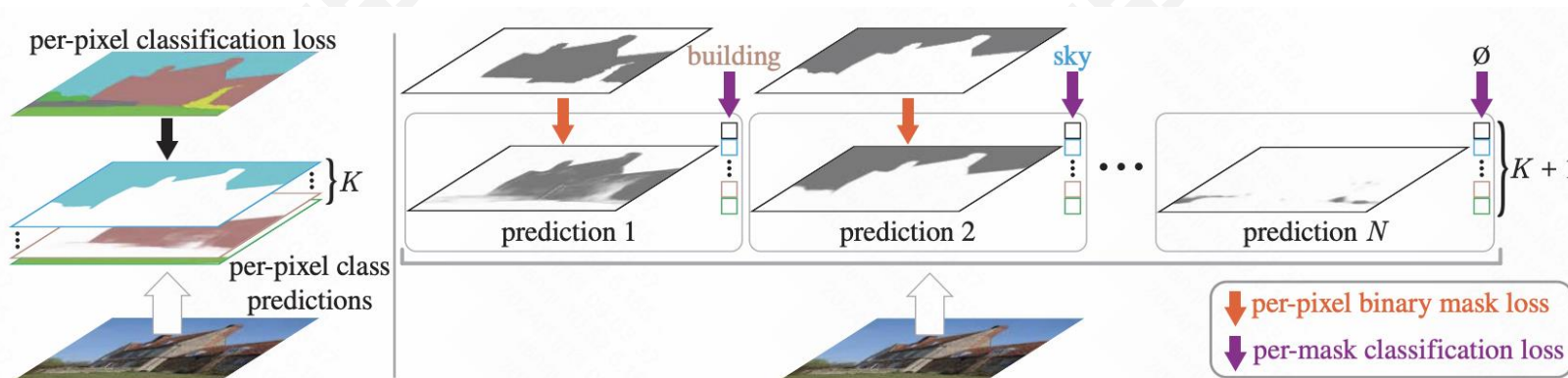
■ Mask R-CNN



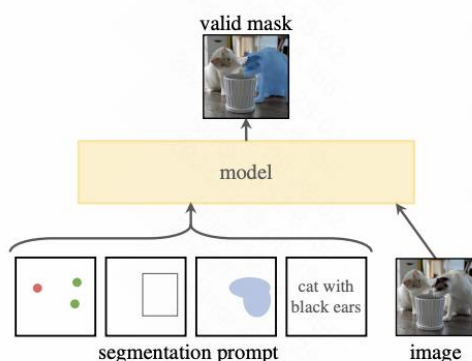
■ Mask R-CNN



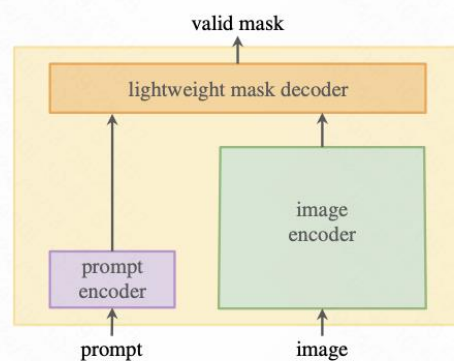
MaskFormer



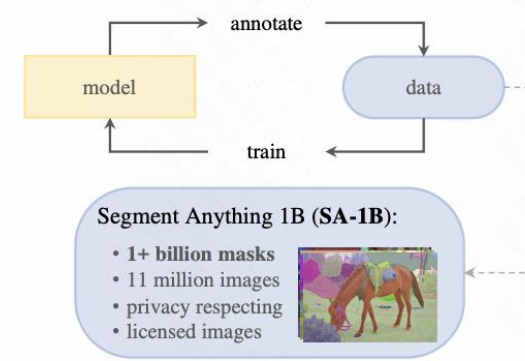
■ Segment Anything Model



(a) **Task:** promptable segmentation



(b) **Model:** Segment Anything Model (SAM)



(c) **Data:** data engine (top) & dataset (bottom)

